

A
PROJECT REPORT
ON
**DESIGN AND SIMULATION OF A FERROELECTRIC CHARGE
PLASMA TUNNEL FIELD EFFECT TRANSISTOR FOR THE
DETECTION OF CHARGED AND UNCHARGED BIOMOLECULES**

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DECLARATION

I hereby declare that the project report entitled -“**Design and simulation of a ferroelectric Charge Plasma Tunnel Field effect transistor for the detection of charged and uncharged biomolecules**”, which is being submitted for the fulfillment of the 7th- 8th semester in Electronics Engineering to Rajkiya Engineering College, Kannauj (UP) is an authentic record of my genuine work done under the guidance of Our Mentor **Mrs. Archana Verma**, Department of Electronics Engineering, Rajkiya Engineering College, Kannauj.

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CERTIFICATE

This is certified that the project entitled “**Design and simulation of a ferroelectric Charge Plasma Tunnel Field effect transistor for the detection of charged and uncharged biomolecules**”, is submitted by Anisha Singh, Yashasvi Singh, Mayank Pal, Sachin Kumar in the partial fulfillment of the award of degree of B. Tech in Electronics Engineering of **Dr. A.P.J. Abdul Kalam Technical University**, is a record student's own work carried under our supervision and guidance. The project report embodies result of original work and studies out by student and the content do not from the award of any other degree to the candidate or to anybody else.

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ABSTRACT

In recent years, the demand for highly sensitive, low-power, and label-free biosensors has increased significantly due to their wide applications in healthcare diagnostics, environmental monitoring, and food safety. Conventional MOSFET based sensors face serious limitations such as high leakage current and poor sensitivity at nanoscale dimensions. To address these challenges, Tunnel Field Effect Transistor (TFET)-based biosensors have emerged as a promising alternative due to their steep subthreshold swing and low power consumption.

In this work, a dielectrically modulated pocket doped TFET biosensor is analyzed for label-free detection of biomolecules. The sensing mechanism is based on variations in dielectric constant and charge density of biomolecules placed in the cavity region. The proposed device demonstrates significant improvement in ON current, sensitivity, and selectivity due to enhanced band-to-band tunneling. Detailed analysis of transfer characteristics, threshold voltage shift, and sensitivity parameters has been carried out. The results indicate that the device exhibits excellent sensitivity for higher dielectric constant biomolecules and is capable of distinguishing between neutral and charged biomolecules effectively. Hence, the proposed biosensor can be considered a strong candidate for next-generation biomedical sensing applications.

In addition, the integration of ferroelectric materials introduces the negative capacitance effect, which enhances gate control and improves the overall device sensitivity. The charge plasma concept eliminates the need for physical doping by inducing carriers using metal work functions.

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CHAPTER 1: INTRODUCTION

1.1 Background and Need for Biosensors

Over the last several decades, the semiconductor industry has been following Moore's Law, where by shrinking the dimensions of CMOS transistor has enabled remarkable progress in terms of speed, integration density, and computational efficiencies; however, due to the ever-increasing size reduction of devices, traditional CMOS circuits face several challenges including increased power consumption, leakage current, and short channel effects. Conduction of current in MOSFETs relies on the thermionic emission model which restricts the subthreshold swing (SS) to around 60 mV/decade at room temperature, thus causing a sharp rise in OFF-state leakage current (I_{OFF}) along with problems such as DIBL when lowering the supply voltage level. Besides these limitations, there has been an increasing demand for high-sensitivity and low-power consuming biosensors used for medical and environmental applications; however, traditional biosensors exhibit a high rate of power dissipation and relatively lower sensitivity. To overcome these difficulties, Tunnel Field Effect Transistors that work based on band-to-band tunneling enable enhanced switching properties and reduced leakage current. In addition, the charge plasma allows for the elimination of doping, while the introduction of ferroelectric material offers enhanced device performance through negative capacitance. [19], [31] Therefore, FE-CP-TFET proves to be an efficient technique to detect charged [17], [34]

1.2 Limitations of Conventional CMOS Technology

Standard complementary metal-oxide-semiconductor (CMOS) technology, despite its extensive utilization in the field of modern electronics, encounters several difficulties when scaling down to the nanometer range. Among the most pronounced problems associated with conventional CMOS is the existence of the theoretical minimum value of subthreshold swing (SS). Being limited to about 60 mV/decade owing to the thermionic-emission-based operation of transistors, the problem with lowering the value of SS becomes an important issue when trying to reduce the supply voltage (V_{DD}) without substantially increasing the OFF-state leakage current (I_{OFF}). Other disadvantages associated with scaling include strong short-channel effects (SCEs) such as drain-induced barrier lowering (DIBL), threshold voltage roll-off, and increased leakage currents. Moreover, high power consumption, heat dissipation, and variations caused by randomness of doping distribution impose certain limitations on the future scaling of CMOS transistors. This prevents further exploitation of the standard CMOS technology in low-power devices like biosensors. [12], [18]

1.3 Tunnel Field Effect Transistor (TFET) for Biosensing

Tunnel TFETs represent an innovative concept in semiconductor technology because of their distinctive mode of operation based on band-to-band tunneling (BTBT) conduction, which helps to surpass the critical limit of SS of standard MOSFETs.

In contrast with MOSFETs, the principle of carrier transport in TFETs is based on tunneling rather than thermionic injection, resulting in higher slope and reduced leakage currents. Thus, TFETs represent a perfect choice for low-power and high-sensitivity electronics. In

biosensing systems, electrical parameters of TFETs are highly sensitive to any external environmental conditions, including the dielectric properties of a medium adjacent to the gate. Upon introduction of biomolecules into the cavity, their interaction with the surface creates changes in the electric field profile and dielectric constant at the surface region, thereby affecting the probability of tunneling process and output current of the transistor. As a result, the electrical parameters of TFETs vary depending on the presence of the analyte molecules. Thanks to their steep slope, extremely low OFF-state currents, and sensitivity, TFETs become a valuable tool in advanced detection of both charged and neutral biomolecules in biosensor devices [15], [17], [32]

1.4 Charge Plasma Concept in TFET

The charge plasma concept in TFET eliminates the need for conventional doping by inducing charge carriers in an intrinsic semiconductor using metals with different work functions. A high work function metal induces holes in the source region, while a low work function metal induces electrons in the drain region, forming p-type and n-type regions without doping. This reduces fabrication complexity and improves device reliability, making it suitable for nanoscale biosensing applications [16], [28], [29]

- **Work Function Engineering:** By selecting appropriate metal work functions, charge carriers are induced in an intrinsic semiconductor without physical doping. A high work function metal at the source induces holes, while a low work function metal at the drain induces electrons, forming p-type and n-type regions. This provides better control over carrier concentration, improves tunneling efficiency, and simplifies fabrication compared to conventional doping methods.[16]
- **Suitability for Biosensing Applications:** The charge plasma concept is well suited for biosensing due to its stable and uniform device characteristics. The absence of doping eliminates variability, resulting in improved reliability and consistent performance. This enables accurate detection of dielectric changes caused by biomolecules, enhancing sensitivity for both charged and uncharged biomolecule detection.[29],[34]

1.5 Ferroelectric Materials and Negative Capacitance Effect

Ferroelectric materials exhibit a unique property known as negative capacitance, which enables internal voltage amplification within the device. When integrated into the gate stack of a TFET, this effect strengthens the electric field at the tunneling junction, thereby increasing the band-to-band tunneling probability. As a result, the device achieves reduced subthreshold swing, improved switching characteristics, and enhanced ON-state current.[19],[30]

In addition, the negative capacitance effect helps in overcoming the conventional 60 mV/decade limitation of MOSFETs, making the device more efficient for low-power operation. This improved electrostatic control also enhances the sensitivity of the device to small variations in the surrounding environment. Therefore, the incorporation of ferroelectric materials significantly improves the overall performance of TFET-based biosensors, making

them highly suitable for detecting both charged and uncharged biomolecules.[5], [34]

1.6 Detection of Charged and Uncharged Biomolecules

Detection of both charged and uncharged biomolecules in TFET-based biosensors depends greatly on the dielectric modulation technique. As explained earlier, a nanocavity is placed at the gate region where the biomolecules reside. The presence of the biomolecules changes the dielectric constant in addition to the electric field distribution. The effect of uncharged biomolecules is limited to changing the dielectric constant of the nanocavity; thus, it causes variation in tunneling probability and drain current. Charged biomolecules affect the TFET by introducing electric charges that cause change in the energy band alignment and therefore produce an appreciable shift in threshold voltage as well as other changes in the current level. The change in these electrical characteristics like drain current and threshold voltage provides means for detecting charged and uncharged biomolecules.[10], [15], [17]

1.7 Objective of the Project

To address the limitations of conventional CMOS-based biosensors, this project presents a comprehensive 2D TCAD investigation of a Ferroelectric Charge Plasma Tunnel Field Effect Transistor (FE-CP-TFET) using the Silvaco ATLAS simulator. The primary objectives are:

- To design and optimize a doping-less TFET structure using the charge plasma concept by selecting appropriate metal work functions for source and drain regions.
- To analyze the effect of ferroelectric materials and dielectric modulation on device performance, focusing on charged and uncharged biomolecules.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to TFET- Based Biosensors

The Figure 2.1 in the “TFET MOSFET.jpg” presents a schematic diagram demonstrating the basic structure and functioning principle of the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) and the Tunnel Field-Effect Transistor (TFET). As shown in the figure, the basic structure of the MOSFET device, which is depicted on the left side, includes the n⁺-type source and drain regions separated by the p-type substrate region to form the n-p-n junction. The working mechanism of this device is based on the concept of thermionic emission, whereby the charge carriers must pass through the physical energy barrier before flowing from the source to the drain regions. On the contrary, the TFET structure on the righthand side adopts a p-i-n (p-type, intrinsic, and n-type) approach. In addition, unlike the MOSFET device that works using thermionic emission, the TFET works using Band-to-Band Tunneling (BTBT). In this case, the electrons can tunnel through the junction barrier. The structural design presented above is relevant to your study since the TFET can have a subthreshold swing below 60mV/decade.

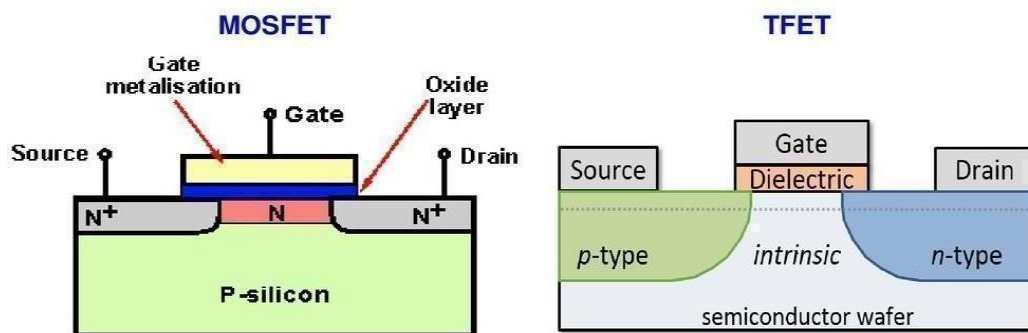


Fig 2.1 structure difference in MOSFET and TFET [26]

2.2 Dielectric Modulation Technique for Biosensing

The Dielectric modulation is a commonly employed technology in TFET-based biosensors. It exploits changes in the local dielectric environment for detection of biomolecules. According to this method, a nano-cavity is placed near the gate of the TFET, and the biomolecules are deposited there. The presence of biomolecules leads to changing of the dielectric constant and consequently the electric field as well as gate capacitance. Thus, probability of band-to-band tunneling increases or decreases, leading to changes in various electrical properties, including drain current, threshold voltage, and subthreshold swing. Uncharged biomolecules affect the dielectric constant of the cavity, while charged ones lead to additional electric charges within the device. Thus, this method allows one to detect biomolecules label-free and in real time. Its high sensitivity makes this method particularly valuable for biosensing applications.

The effect of uncharged biomolecules consists of changes in the dielectric constant, which causes relatively mild changes in electrical properties. However, charged biomolecules provide additional effects, due to the introduction of electric charges, changing the electric field and thus altering the energy bands, leading to significant shifts in the current and threshold voltage. Sensitivity of this method depends on the cavity size, dielectric constant of the molecules, and some other

2.3 Charge Plasma Based Devices

Explanation of the Figure: 2.2 This diagram illustrates the transition from traditional semiconductor doping to the Charge Plasma (CP) concept in Tunnel FETs. Structure (a) - Conventional DG-TFET: This represents a standard Double Gate TFET where the p^+ source and n^+ drain are typically formed using physical doping (like ion implantation). It uses a Control Gate (CG) to modulate the intrinsic channel. Structure (b) - Charge Plasma TFET: In this architecture, the semiconductor layer remains Intrinsic (I-Si) throughout. The source and drain regions are created "electrically" by placing metal electrodes (Ni-Si) with specific work functions (labelled PG-1 and PG-2) near the silicon. These electrodes induce a high concentration of holes or electrons at the surface, effectively creating p^+ and n^+ regions without the need for high-temperature doping processes. The L_{GAP} labels indicate the spacing between the gates, which is a critical design parameter you likely analyzed in your Silvaco simulations to minimize leakage current and optimize the tunneling volume.

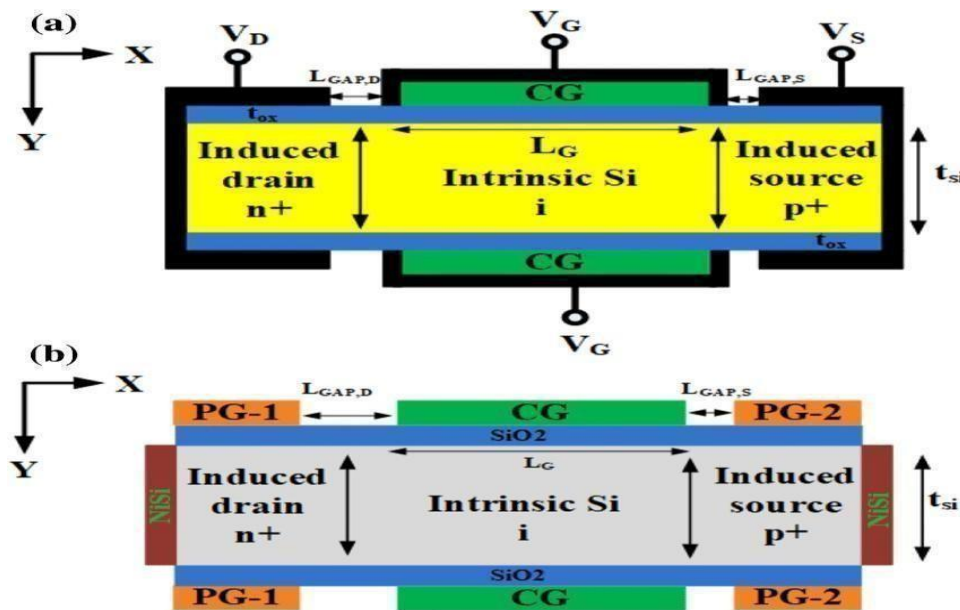


Fig 2.2 structure difference in MOSFET and TFET [21]

2.4 Ferroelectric Materials

The surface potential profile along the lateral distance of the proposed FE-CP-TFET is a critical indicator of its electrostatic response to biological analytes. As illustrated in Fig. 2.3, the surface potential remains at a baseline level when the nano-cavity is filled with air ($K=1$). However, the introduction of biomolecules with higher dielectric constants ($K=5$, 7 , and 10) leads to a prominent localized increase in the surface potential, particularly at the source-channel tunneling junction. This enhancement is primarily driven by the improved capacitive coupling and the internal voltage amplification provided by the ferroelectric negative capacitance layer. The resulting surge in surface potential aggressively pulls down the energy bands, narrowing the tunneling barrier width and significantly increasing the drive current for high-sensitivity detection. This localized potential modulation confirms that the device is highly effective as a label-free transducer for detecting variations in biological permittivity.

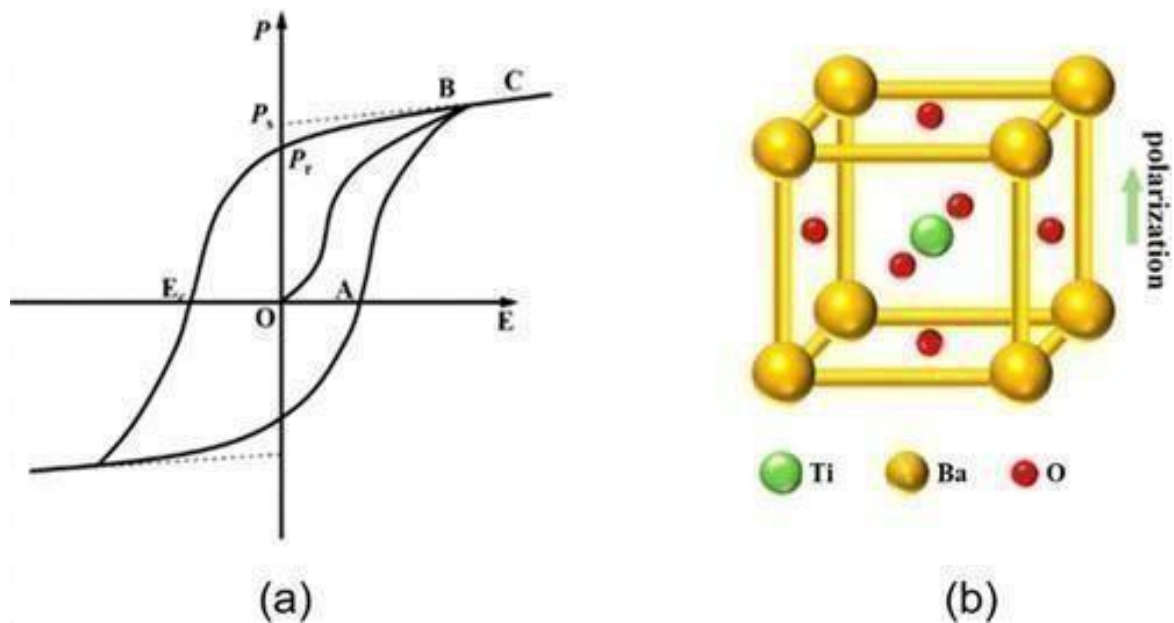


Fig 2.4 Ferroelectric Materials [27]

2.5 Detection Mechanism of Charged vs Neutral Biomolecules

The Changes in the local electrostatic environment and dielectric properties of the medium around the sensing region play an important role in detecting charged and neutral biomolecules using TFETs-based biosensors. In particular, when introduced into the nanocavity near the gate, biomolecules cause some changes in the dielectric constant of the medium, and also influence the distribution of the electric field in the vicinity of the channel. As a rule, neutral molecules affect the dielectric constant, changing the gate capacitance and somewhat changing the band-to-band tunneling probability, resulting in minor variations of the drain current.

Charged molecules have an impact on the electric field and channel potential, altering the energy band alignment, which causes significant changes in threshold voltage and drain current. In particular, introducing positive charges into the nanocavity usually results in increasing electric field intensity, and therefore tunneling current, whereas negative charges may cause decrease in the tunneling probability or other changes in the behavior of the device. Sensitivity of the device depends on several parameters including the charge density, dielectric constant of the biomolecules, dimensions of the nano-cavity, and device geometry.

An analysis of electrical characteristics variations can be used to detect and classify the introduced biomolecules as charged.

2.6 Research Gap

Despite significant advancements in TFET-based biosensors, several challenges still limit their practical performance and widespread application. Conventional TFET designs often rely on chemical doping, which introduces issues such as random dopant fluctuations, increased variability, and fabrication complexity, especially at nanoscale dimensions. Although dielectric modulation techniques have improved biomolecule detection, the sensitivity and stability of these devices remain limited when detecting low-concentration or neutral biomolecules. Furthermore, many existing designs do not effectively utilize advanced concepts such as charge plasma and ferroelectric materials simultaneously, which can significantly enhance device performance. The lack of integrated approaches that combine doping-less structures with negative capacitance effects results in suboptimal control over electric field and tunneling efficiency. Therefore, there is a need to develop an optimized device architecture, such as the Ferroelectric Charge Plasma TFET (FE-CP-TFET), that can overcome these limitations and provide improved sensitivity, low power consumption, and reliable detection of both charged and uncharged biomolecules.

2.7 Proposed Work: FE-CP-TFET

In this work, a Ferroelectric Here, a FE-CP-TFET structure is introduced for the detection of charged and uncharged biomolecules in an energy-efficient manner. In the present structure, the charge plasma approach is employed for the removal of conventional doping from the device structure, in which the carriers are generated

via proper choice of metal work functions that simplify the fabrication process, minimize the variation, and enhance the device uniformity due to nanoscale dimensions. Moreover, the use of ferroelectric material in the structure results in the introduction of negative capacitance

effect, which increases the voltage gain, increases the strength of the electric field, and increases the efficiency of band-to-band tunneling process.

The performance of the introduced structure is analyzed by calculating the important electrical parameters such as drain current, threshold voltage, subthreshold swing, and the ratio of I_{on}/I_{off} under various sensing conditions. Moreover, a parametric analysis is performed to analyze the impact of the dielectric constant, biomolecular charge, and the geometry of the structure on sensitivity and detection ability of the device. For the simulation of the whole structure, Silvaco ATLAS TCAD is used.

CHAPTER 3

FERROELECTRIC CHARGE

PLASMA TFET

FOR BIOSENSING

3.1 Working Principle of TFET

The Fig. 3.1 Modulation of Energy Bands by Biomolecule Charge. The sensing mechanism of the proposed FE-CP-TFET is fundamentally grounded in the modulation of the energy band diagram at the source-channel tunneling junction. Unlike conventional MOSFETs where thermionic emission is the dominant carrier transport, TFETs rely on quantum mechanical Band-to-Band Tunneling (BTBT). As shown in Fig. 3.1 the Reference State represents the device without biomolecules, where a specific tunneling barrier width $\lambda_{t,ref}$ exists. In Detection State A, the presence of negatively charged biomolecules (like DNA) induces an electrostatic repulsion that uplifts the channel potential ($E_{c,neg}$), thereby increasing the tunneling barrier width and exponentially decreasing the tunneling current. Conversely, in Detection State B, positively charged biomolecules lower the channel potential ($E_{c,pos}$), which enhances the electric field at the junction ($E_{s,pos}$) and shrinks the tunneling barrier width ($\lambda_{t,pos} < \lambda_{t,ref}$). This results in an exponential increase in BTBT current, providing a clear electrical signature for the detection of charged analytes. The integrated ferroelectric layer amplifies these shifts through the negative capacitance effect, significantly improving the sensitivity of the energy band modulation.

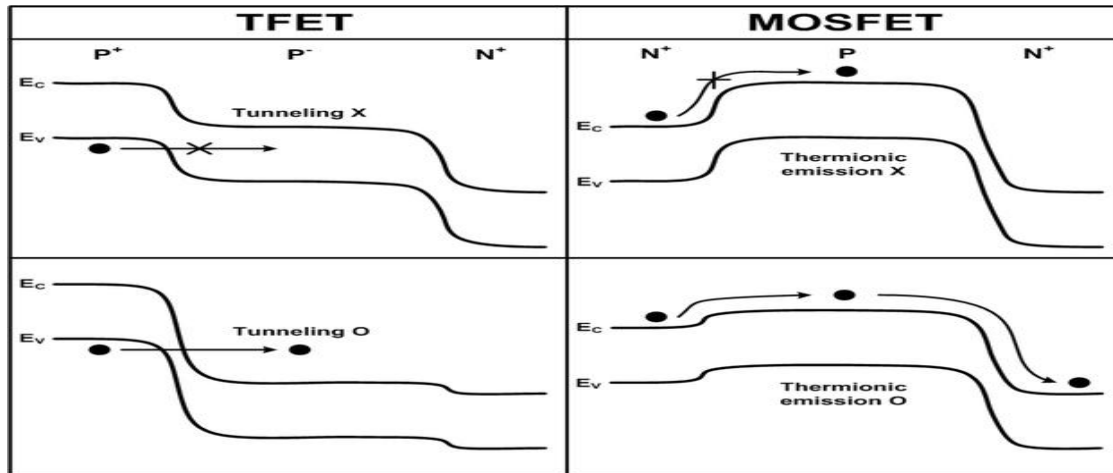


Fig 3.1 structure difference in MOSFET and TFET [30]

3.2 Charge Plasma TFET (CP-TFET) Concept

3.2.1 source (Charge Plasma)

Fig. 3.2.1 Overview of the Biosensing System Workflow and Signal Transduction The operational architecture of a modern biosensing platform is a multi-stage process that integrates biological recognition with advanced electronic signal processing, as depicted in the systemic workflow of Fig. 3.2.1. The process initiates at the Detector stage, where a complex analyte such as DNA, enzymes, or specific proteins—is introduced to the sensing area. In the proposed FE-CP-TFET, the traditional reliance on physical doping is eliminated through the Charge Plasma concept. By utilizing metals with specific work functions, carriers are induced electrostatically in an intrinsic silicon channel, creating a pristine electronic environment that is highly sensitive to external perturbations. Once a biological recognition event occurs, the Transducer stage serves as the bridge that converts microscopic molecular interactions into quantifiable electrical data. The Charge Plasma architecture plays a pivotal role here by providing an ultra-low baseline OFF-state current, typically in the range of 10^{-17} A. This stable baseline ensures that any localized change in the dielectric constant or charge density caused by the biomolecules in the nano-cavity is translated into a massive shift in current or threshold voltage without being lost in parasitic noise. The final phase is the Output System, which involves the read-out technology and signal amplification. The synergy between the Charge Plasma induced source/drain regions and the ferroelectric gate stack provides an inherent internal voltage gain. This means the "Biological Signature" of the analyte is amplified naturally by the device physics, allowing the data acquisition system to process and display high-fidelity results. This end-to-end integration highlights the efficiency of the FE-CP-TFET as a superior, low-power platform for real-time label-free biosensing.

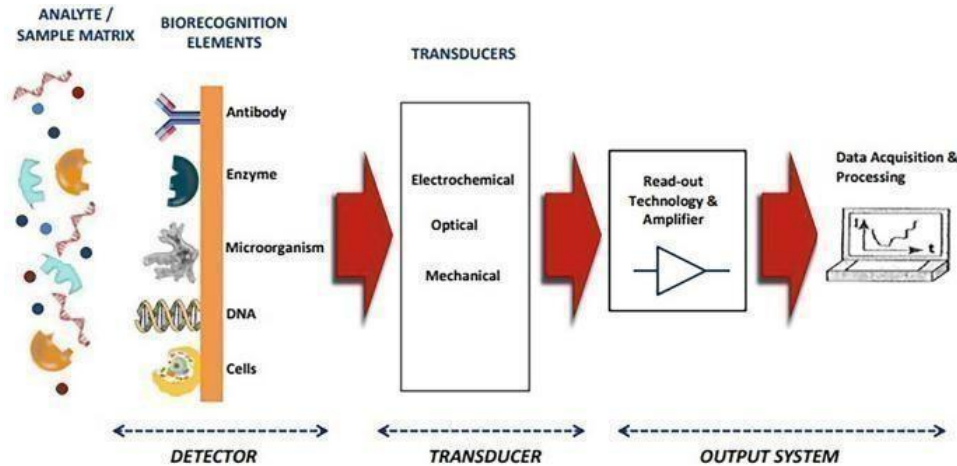


Fig 3.2.1 structure of Charge Plasma [3]

3.2.2 Channel (Intrinsic Silicon)

The channel region Intrinsic silicon makes up the channel region of the proposed FE-CP-TFET, allowing it to act as a medium that transports carriers from the source to the drain. In contrast to conventional transistors, the channel region in this case does not have any doping, avoiding problems like impurity scattering, random dopant fluctuations, and variations at the nanometer level. This helps increase

the reliability and uniformity of the device.

The intrinsic nature of the channel also allows for electrostatic tuning of the energy bands since it increases control over the channel by the gate. When the gate voltage is applied, the electric field modifies the energy bands of the source and the channel region, leading to efficient band-to-band tunneling. The electric field is an important determinant of the tunneling current and thus the intrinsic channel plays an important role in the device.

The intrinsic silicon channel is highly sensitive to changes in the surrounding dielectric properties, including the presence of a nano-cavity that would be used for the detection of biomolecules. Dielectric constant and charges in the surrounding effect the channel potential and, consequently, the tunneling probability. In effect, an intrinsic channel allows for biosensing due to the sensitivity to the surrounding charge.

3.2.3 Gate with Ferroelectric Layer, Oxide Layer, and Nano-Cavity

The gate However, the proposed gate structure is critical for defining the performance parameters of the FE-CP-TFET since it incorporates a metal gate together with ferroelectric HfO_2 layer. By virtue of the presence of ferroelectric material, the phenomenon of negative capacitance occurs within this proposed FE-CP-TFET. As such, there is an increase in the internal voltage amplification capability and consequently the electric field across the channel region. The enhanced electric field boosts the probability of band-to-band tunneling and hence contributes to effective switching performance with a smaller subthreshold swing.

It should be noted that there is an oxide layer between the gate and channel. Common oxide materials include SiO_2 or even high-k dielectric such as HfO_2 . The oxide material insulates the gate from the channel at the same time ensuring good capacitive coupling in order to enable good electrical manipulation of the potential of the channel.

Additionally, there is a nano-cavity at the gate region that acts as the sensing element where biomolecules attach themselves. Usually, the nano-cavity will either contain air or biomolecules depending on whether the FE-CP-TFET sensor is used in sensing biomolecules or not. Depending on the type of biomolecules attached to the nanocavity, there will be different effects.

3.3 Ferroelectric Negative Capacitance Effect

The negative capacitance effect is a unique phenomenon observed in ferroelectric materials, where the polarization response of the material leads to a temporary region of negative differential capacitance. This behavior arises from the internal energy profile of ferroelectric materials, which typically exhibit a double-well energy landscape. In this region, a small increase in charge results in a reduction of the potential energy, effectively producing voltage amplification within the device.

When a ferroelectric layer such as hafnium oxide (HfO_2) is integrated into the gate stack of a

TFET, it operates in series with the conventional oxide layer. This combination stabilizes the negative capacitance region and enhances the overall gate capacitance. As a result, the surface potential in the channel increases more rapidly than the applied gate voltage, leading to stronger electrostatic control over the channel.

This enhanced control significantly improves the electric field at the source-channel junction, which is critical for band-to-band tunneling. The increased electric field reduces the tunneling barrier width, thereby increasing the tunneling current and improving the ON-state performance of the device. At the same time, the subthreshold swing is reduced below the conventional limit of 60 mV/decade, enabling ultra-low power operation.

Additionally, the negative capacitance effect improves the sensitivity of the device to small perturbations in the surrounding dielectric environment. In biosensing applications, even minor changes caused by biomolecules in the nano-cavity can lead to noticeable variations in electrical characteristics such as drain current and threshold voltage. This makes the FE-CP-TFET highly effective for detecting both charged and uncharged biomolecules with high precision.

Overall, the incorporation of the ferroelectric negative capacitance effect provides improved switching efficiency, enhanced sensitivity, reduced power consumption, and better scalability, making it a key feature in next-generation TFET-based biosensors.

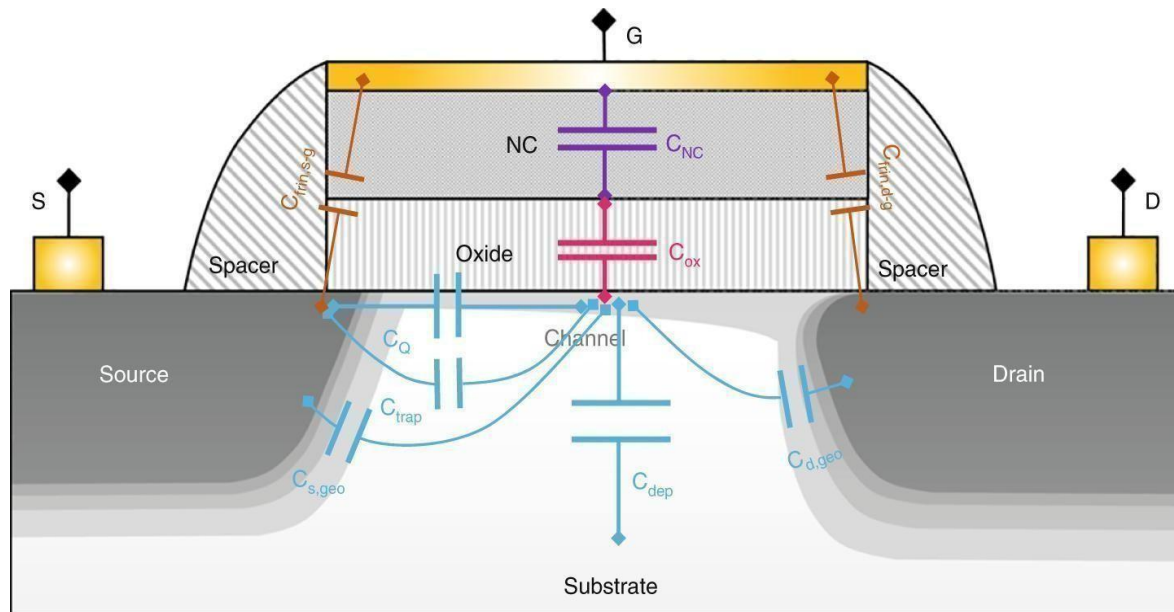


Fig 3.3 Ferroelectric Negative Capacitance Effect [34]

3.4 Proposed FE-CP-TFET Structure

The proposed Ferroelectric Charge Plasma Tunnel Field Effect Transistor (FE-CP-TFET) is an advanced nanoscale device designed to achieve high sensitivity and ultra-low power operation for biosensing applications. This structure integrates three key concepts: tunnel field-effect operation, charge plasma carrier formation, and ferroelectric-based negative capacitance

enhancement. The combination of these mechanisms significantly improves the electrical characteristics and sensing capability of the device compared to conventional transistor-based biosensors.

3.4.1 Overview of Device Architecture

The FE-CP-TFET structure is based on an intrinsic (undoped) silicon channel, eliminating the need for conventional doping processes. Instead of introducing impurities, carrier regions are formed using the charge plasma concept, where metals with appropriate work functions are used to induce electron and hole concentrations in the source and drain regions.

The basic structure consists of:

1. Source region (p-type induced)
2. Drain region (n-type induced)
3. Gate dielectric layer
4. Ferroelectric layer integrated with the gate
5. Ferroelectric layer integrated with the gate

3.4.2 Source and Drain Formation Using Charge Plasma

Unlike conventional TFETs that rely on heavy doping to form p-n junctions, the proposed structure uses metals with specific work functions to induce charge carriers in an intrinsic semiconductor. This phenomenon is known as charge plasma formation.

This approach offers several advantages:

1. Eliminates ion implantation and diffusion steps.
2. Reduces random dopant fluctuations.
3. Improves device reliability and uniformity.
4. Enables better electrostatic control at nanoscale dimensions.

3.4.3 Channel Region

The channel is made of intrinsic silicon, which serves as the tunneling region between the source and drain. The absence of doping ensures:

1. Reduced scattering effects
2. Lower leakage currents
3. Improved subthreshold characteristics

3.4.4 Gate Stack with Ferroelectric Layer

One of the most important features of the proposed structure is the integration of a ferroelectric material in the gate stack. The gate structure typically consists of:

1. Metal gate electrode

2. Ferroelectric layer
3. Conventional dielectric layer (such as SiO₂ or HfO₂)

3.5 Proposed FE-CP-TFET Structure

- 1 The proposed Ferroelectric Charge Plasma Tunnel Field Effect Transistor (FE-CP-TFET) is a breakthrough innovation in the area of bio-sensing using semiconductors. The use of the concepts of tunneling switch, charge plasma carriers and ferroelectric negative capacitance allow the sensor to address several shortcomings that exist in other traditional bio-sensors based on either MOSFET or even TFET. Such advances have made the FE-CP-TFET very suitable for today's biomedical and environmental bio-monitoring purposes due to its high sensitivity and low power consumption.
- 2 One of the most prominent advantages of the FE-CP-TFET is its ability to achieve ultra-high sensitivity. Sensitivity is one of the key factors that define the performance of biosensors because it represents the lowest concentration level of the target biomolecule that can be detected. In the developed design, the sensitivity is greatly increased due to the high electrostatic interaction between the gate and the channel, which is strengthened by the ferroelectric layer. The negative capacitance property of the ferroelectric component enables the increase of the electric field within the device without increasing the applied gate voltage. Thus, even a small disturbance in the vicinity of the nano-cavity, such as the presence of the biomolecule, will cause a measurable change in the drain current.
- 3 Another major advantage of The ability of the FE-CP-TFET to identify both charged and neutral biomolecules efficiently constitutes another advantage of the device. In normal biosensors, charged biomolecules can only be detected since their charges affect the electrostatic potential present within the sensor. However, in the device presented above, the process of detection involves the use of dielectric modulation through the nano-cavity. Thus, the presence of a neutral biomolecule within the cavity will lead to changes in the dielectric constant of the medium, thus altering the capacitance of the gate and the electric field distribution in the channel region. The resultant effect will affect the tunneling probability, hence the drain current. On the other hand, charged biomolecules will have an even greater impact on the performance of the device since the charges will act upon the potential of the channel region, thereby shifting the threshold voltage.
- 4 The improvement in subthreshold swing is another critical advantage of the FE-CP-TFET. In conventional MOSFET devices, the subthreshold swing is fundamentally limited to 60 mV per decade at room temperature, which restricts the switching efficiency and sensitivity. However, the proposed device leverages both the tunneling mechanism of TFETs and the negative capacitance effect of the

ferroelectric layer to achieve a much steeper subthreshold slope. This results in faster switching and a more pronounced change in current for a given change in gate voltage. A steeper subthreshold swing enhances the device's ability to detect small variations in the presence of biomolecules, thereby improving its overall performance as a biosensor. Additionally, this feature allows the device to operate at lower supply voltages, which is highly desirable in portable and battery-operated systems.

- 5 Low power consumption is another significant benefit offered by the FE-CP-TFET structure. The combination of steep switching characteristics and reduced leakage current ensures that the device consumes minimal power during operation. The intrinsic channel used in the charge plasma concept minimizes leakage pathways, while the tunneling-based conduction mechanism reduces the dependence on high thermal energy. Furthermore, The effect of the negative capacitance effect results in a reduction of the amount of power needed because it requires reduced gate voltage. Therefore, this technology can be used in various applications that need high levels of power efficiency, including wearable health devices and implanted medical devices.
- 6 The elimination of conventional doping processes through the charge plasma concept is also a noteworthy advantage. In traditional semiconductor fabrication, doping is achieved through ion implantation and diffusion, which introduce impurities into the material. These processes not only increase fabrication complexity but also lead to variability due to random dopant fluctuations, especially at nanoscale dimensions. In the FE-CP-TFET, carriers are induced in the intrinsic silicon using metals with appropriate work functions, thereby eliminating the need for doping. This results in a more uniform device structure, improved reproducibility, and reduced manufacturing complexity. Additionally, the absence of doping-related defects enhances the reliability and longevity of the device.
- 7 The FE-CP-TFET also demonstrates an improved ON/OFF current ratio, which is essential for accurate signal detection in biosensing applications. A high ON-current ensures that the device produces a strong signal when biomolecules are present, while a low OFF-current minimizes background noise when no biomolecules are detected. The tunneling mechanism, combined with enhanced gate control due to the ferroelectric layer, contributes to a high ON-current. At the same time, the intrinsic channel and absence of doping reduce leakage currents, resulting in a very low OFF-current. This high I_{on}/I_{off} ratio improves the signal-to-noise ratio, making it easier to distinguish between different sensing conditions and reducing the likelihood of false positives or negatives.
- 8 Another important advantage of the proposed device is its ability to perform label-free detection. In many traditional biosensing techniques, biomolecules must be labeled with fluorescent or radioactive tags to enable detection. These labeling processes are often time-consuming, expensive, and may alter the natural properties of the biomolecules. In contrast, the FE-CP-TFET detects biomolecules directly based on their intrinsic electrical properties, such as charge and dielectric constant. This eliminates the need for additional processing steps and allows for real-time monitoring of biological interactions. Label-free detection not only simplifies the

sensing process but also makes it more cost-effective and suitable for large-scale deployment.

- 9 The scalability and compatibility of the FE-CP-TFET with existing semiconductor technologies further enhance its practical applicability. The device structure can be fabricated using standard CMOS-compatible processes, making it easier to integrate with existing electronic systems. Its nanoscale dimensions allow for high-density integration, enabling the development of compact and multifunctional biosensing platforms. This is particularly advantageous for lab-on-chip applications, where multiple sensing elements can be integrated onto a single chip to perform simultaneous detection of various analytes.
- 10 Reliability and stability are also improved in the proposed structure due to the absence of doping and the use of stable ferroelectric materials. The reduced variability in device parameters ensures consistent performance across multiple devices, which is crucial for commercial applications. Moreover, the use of robust materials and simplified fabrication processes enhances the durability of the device, allowing it to operate effectively over extended periods and under varying environmental conditions

CHAPTER 4

DEVICE STRUCTURE AND

SIMULATION

METHODOLOGY

4.1 Device Structure and Design

Figure 4.1 shows the detailed two-dimensional structure of the proposed Ferroelectric Charge Plasma Tunnel Field Effect Transistor (FE-CP-TFET) biosensor developed for ultra-sensitive and low-power biomedical sensing applications. The device is fabricated on an intrinsic silicon body with a silicon thickness (T_{si}) of 10 nm, which helps eliminate random dopant fluctuations and improves the reliability of nanoscale operation. The channel region is divided into different sections, including the source-side region ($L_{g1} = 0.02 \mu m$), central channel region ($L_{g2} = 0.4 \mu m$), and drain-side region ($L_{g3} = 0.02 \mu m$), providing strong electrostatic control over carrier transport.

As compared to the traditional TFET architecture, the formation of the source and drain does not involve physical doping, rather a charge plasma technology is employed for this purpose. The use of high work function metal ($WFS = 5.93 \text{ eV}$) on the source end results in a p-type hole plasma region, whereas low work function metal ($WFD = 3.9 \text{ eV}$) on the drain end causes an n-type electron plasma region to be created.

The biosensing mechanism mainly takes place within the region of nano-cavity present below the gate. This cavity has a size of ($L_{cavity} = 0.08 \mu m$) in terms of length, while the thickness is ($T_{cavity} = 8 \text{ nm}$). In this cavity, the biomolecules get fixed to be sensed by the device. The variation in dielectric constant due to the presence of biomolecules directly affects the channel potential as well as tunneling probability.

The gate stack consists of both top and bottom gates with a gate work function of 4.8 eV, forming a dual-gate configuration that provides superior gate controllability and enhanced electrostatic coupling. A thin ferroelectric HfO_2 layer with thickness ($T_{HfO_2} = 0.5 \text{ nm}$) is integrated into the gate oxide structure to introduce the negative capacitance effect. This ferroelectric amplification internally boosts the applied gate voltage, resulting in stronger band bending, reduced tunneling barrier width, and improved band-to-band tunneling efficiency at ultra-low operating voltages.

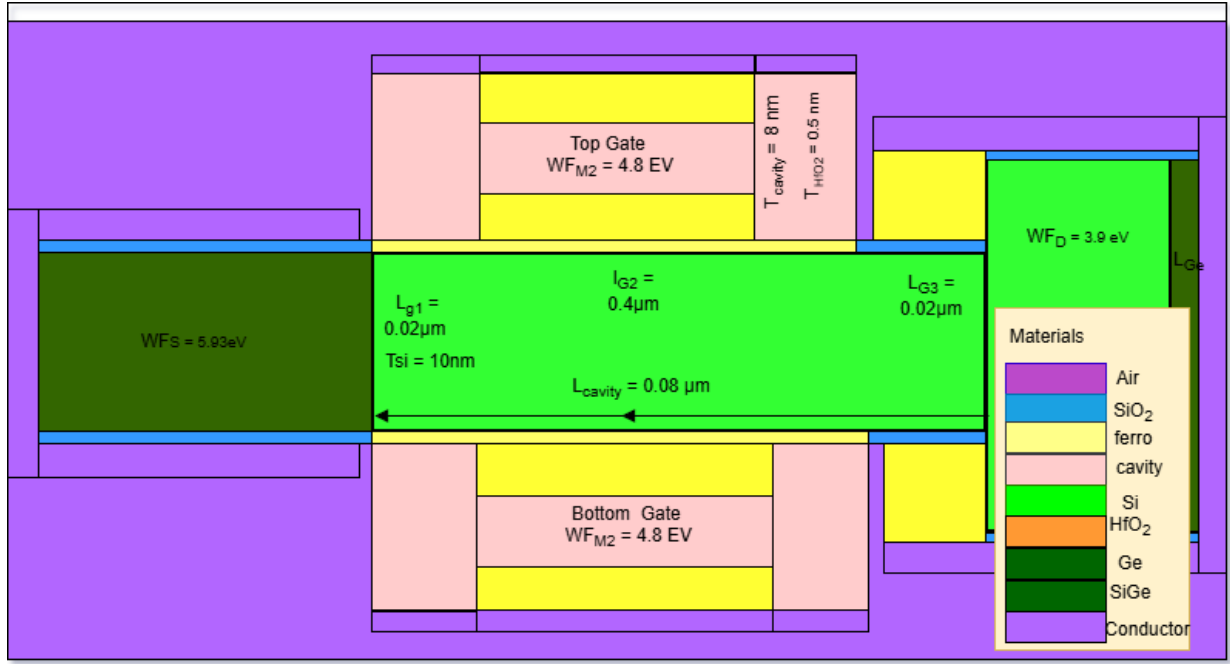


Fig 4.1 2D structure of the proposed FE-CP-TFET biosensor

4.2 Nano-Cavity Formation for Biomolecule Detection

In the proposed FE-CP-TFET structure, a nano-cavity is introduced near the gate region to enable biomolecule detection through dielectric modulation. This nano-cavity acts as the sensing region where biomolecules are immobilized or captured. It is typically formed by creating a small gap or etched region in the oxide layer close to the channel, which can be filled with air or biomolecules.

The existence of biomolecules within the nano-cavity modifies the value of the dielectric constant in that area. Such modification results in changes in the distribution of the electric field and gate capacitance, which ultimately affect the band-to-band tunneling of charge carriers in TFETs. Ultimately, variations are observed in the electric parameters like drain current, threshold voltage, and subthreshold slope.

In the case of uncharged molecules, the key influence lies in the altered dielectric constant, which affects the electrical field and tunneling mechanism. Conversely, charged molecules add another electric charge, which causes notable effects on the channel voltage and energy band structure.

The size, position, The dimensions, location, and dielectric behavior of the nano-cavity have a significant influence on the performance of sensitivity of the biosensor. A carefully engineered nano-cavity boosts the coupling between the biomolecule and the channel region, thus increasing the sensitivity. Hence, having a nano-cavity in the biosensor constitutes one of the most important characteristics of the FE-CP-TFET-based biosensor.

4.3 Role of Ferroelectric Layer

In this regard, the ferroelectric layer acts as an important contributor towards the improved working of the suggested FE-CP-TFET. Generally, it is included in the gate stack structure and comprises materials like hafnium oxide (HfO_2), which are used in current semiconductors. The main task of this layer is that it brings the phenomenon of negative capacitance and generates the voltage gain internally.

In view of this, the strength of the electric field within the channel region increases greatly, more specifically at the source-channel boundary, due to the phenomenon of band-to-band tunneling. The high strength of the electric field makes it possible to minimize the width of the tunneling barrier, which leads to an increase in tunneling probability and high ON state current. It also helps reduce the subthreshold swing.

The introduction of the ferroelectric layer also leads to an improved electrostatic gating of the gate with respect to the channel. It facilitates the modulation of the channel potential more efficiently while increasing its sensitivity. When applied for biosensors, slight changes in the dielectric medium due to the presence of biological molecules become easily detectable because of the improved gating effect.

Overall, the incorporation of the ferroelectric layer significantly enhances device efficiency, reduces power consumption, and increases sensitivity, making the FE-CP-TFET highly suitable for detecting charged and uncharged biomolecules.

4.4 Simulation Tool (Silvaco ATLAS)

The proposed FE-CP-TFET device is designed and analyzed using the Silvaco ATLAS TCAD simulator, which is a powerful tool for modeling and simulation of semiconductor devices at nanoscale dimensions. ATLAS provides a flexible environment to define complex device geometries, material properties, and boundary conditions, enabling accurate prediction of device behavior under various operating conditions. It is particularly suitable for TFET-based devices due to its ability to model advanced physical phenomena such as band-to-band tunneling and carrier transport mechanisms.

In this work, the device structure is carefully constructed by defining multiple regions such as the source, drain, intrinsic silicon channel, oxide layer, ferroelectric layer, and nano-cavity. Each region is assigned appropriate material parameters, including dielectric constant, electron affinity, bandgap, and thickness. The charge plasma concept is implemented by selecting suitable metal work functions to induce hole and electron carriers in the source and drain regions, respectively, without the need for conventional doping.

To accurately capture the device physics, several important physical models are incorporated into the simulation. These include band-to-band tunneling (BTBT) models for current generation, Shockley–Read–Hall (SRH) recombination for carrier loss mechanisms, and mobility models to describe carrier transport. In addition, the simulation considers electric field distribution, energy band variation, and charge concentration profiles within the device.

The simulation is performed under different biasing conditions by varying gate voltage and

drain voltage to obtain the electrical characteristics of the device. Key output parameters such as drain current (I_d), threshold voltage (V_{th}), subthreshold swing (SS), and I_{on}/I_{off} ratio are extracted and analyzed. These parameters are crucial for evaluating the performance of the device in biosensing applications.

Furthermore, the nano-cavity region is modeled by assigning different dielectric constants and charge conditions to represent various biomolecules. The impact of these biomolecules on the device characteristics is studied by observing changes in current and voltage parameters. This approach allows the evaluation of device sensitivity and detection capability for both charged and uncharged biomolecules.

Overall, the Silvaco ATLAS simulation framework provides a comprehensive platform for designing, analyzing, and optimizing the proposed FE-CP-TFET, ensuring accurate performance evaluation and effective implementation for advanced biosensing applications.

4.5 Temperature and Dielectric Influence on Charge Plasma TFET Biosensors for Biomedical Applications

Temperature and dielectric properties play an important role in the performance of Charge Plasma TFET biosensors used for biomedical applications. In these biosensors, biomolecules are detected based on the changes they produce in the electrical characteristics of the device. Temperature affects carrier concentration, bandgap energy, and tunneling current in the TFET structure. As temperature increases, leakage current and thermal noise also increase, which may reduce the sensing accuracy and stability of the biosensor. On the other hand, the dielectric constant of biomolecules strongly influences the electric field and surface potential inside the sensing cavity. Biomolecules with higher dielectric constants improve gate control and increase band-to-band tunneling probability, resulting in higher drain current and better sensitivity. The charge plasma concept eliminates the need for heavy doping, reducing fabrication complexity and improving device reliability. Additionally, the incorporation of ferroelectric materials in FE-CE-TFET structures enhances internal electric field amplification and provides improved sensitivity with ultra-low power operation. Due to these advantages, FE-CE-TFET biosensors are highly suitable for biomedical applications such as DNA detection, cancer diagnosis, protein sensing, and glucose monitoring.

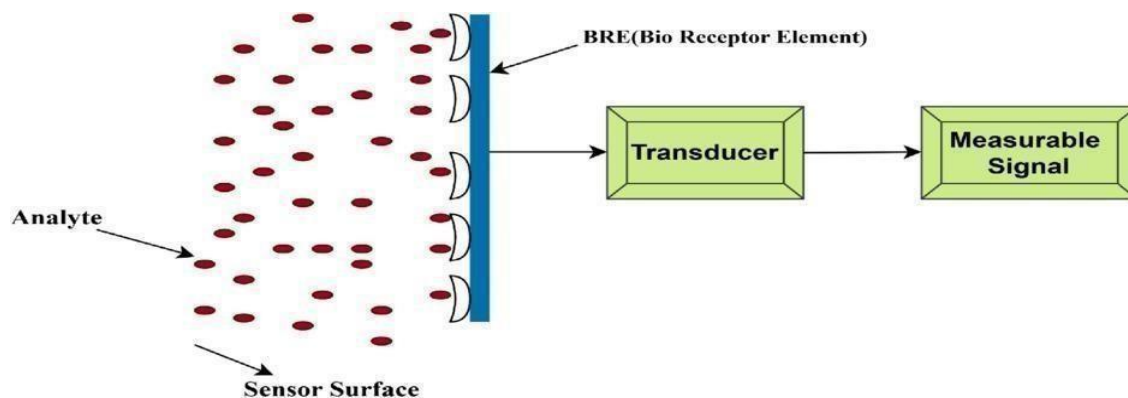


Fig 4.5 Schematic diagram of key component of biosensor [2]

The figure represents the basic working principle of a biosensor used in biomedical sensing applications. In this diagram, the small red particles represent the analyte, which may be biomolecules such as DNA, proteins, glucose, or viruses present near the sensor surface. The blue vertical layer represents the BRE (Bio-Receptor Element), which is responsible for specifically recognizing and binding the target biomolecules. When the analyte interacts with the bio-receptor element, a biological reaction takes place on the sensor surface. This interaction is then detected by the transducer, which converts the biological response into an electrical signal. Finally, the output is obtained in the form of a measurable signal, such as current, voltage, or frequency change, which can be analyzed for biomedical diagnosis. In FE-CE-TFET biosensors, the transducer section is based on the TFET structure, where the presence of biomolecules changes the dielectric environment and affects the tunneling current, enabling highly sensitive detection.

4.6 Device Parameters and Model Setup

The performance of the proposed FE-CP-TFET is strongly dependent on the selection of device parameters and the accuracy of the physical models used in the simulation. In this work, the device is designed using optimized geometrical and material parameters to ensure efficient operation for biosensing applications.

The channel region is made of intrinsic silicon with a carefully chosen channel length to maintain proper electrostatic control and enhance tunneling efficiency. The thickness of the oxide layer, typically silicon dioxide (SiO_2) or high-k dielectric such as hafnium oxide (HfO_2), is selected to achieve strong gate control while minimizing leakage. The ferroelectric layer thickness is also optimized to ensure effective negative capacitance behavior and improved electric field amplification.

The source and drain regions are formed using the charge plasma concept, where appropriate metal work functions are assigned to induce hole and electron carriers. The work function values are carefully selected to create sufficient carrier concentration and maintain proper energy band alignment for efficient band-to-band tunneling.

In addition to structural parameters, several physical models are enabled in the Silvaco ATLAS simulator to accurately capture device behavior. These include band-to-band tunneling (BTBT) models for current generation, Shockley–Read–Hall (SRH) recombination for carrier lifetime effects, and mobility models for carrier transport. These models ensure realistic simulation of electrical characteristics.

4.7 Biomolecule Parameters (Dielectric Constant and Charge)

4.7.1 Geometrical and Material Parameters (Table 1):

The suggested FE-CP-TFET biosensor is fabricated based on intrinsic Si channel with silicon body thickness (T_{si}) of 10 nm and channel length (L_{ch}) of 40 nm. The intrinsic silicon channel proves to be very useful for several reasons; namely, it prevents dopant fluctuations, leading to increased reliability, stability, and scalability for nanoscale applications.

The source and drain segments have the dimensions of 20 nm each and were created using the charge plasma approach without employing chemical doping. The charge plasma approach involves applying metals that possess appropriate work functions to the silicon body.

induce an n-type electron plasma. This doping-less structure reduces impurity scattering effects and improves the electrical performance of the device.

The biosensing operation is mainly performed through the nanogap cavity region, which has a cavity length (L_{cav}) of 10 nm and a cavity thickness (T_{cav}) of 5 nm. Biomolecules are immobilized inside this cavity, and their dielectric properties modulate the local electric field and channel potential of the device. The gate stack is further enhanced using a 10 nm thick ferroelectric layer based on Si: HfO₂ and a 2 nm silicon dioxide (SiO₂) interfacial oxide layer. The ferroelectric layer provides negative capacitance behavior, which internally amplifies the applied gate voltage and improves switching performance.

Table 1: Geometrical & Material Parameters

Parameter	Symbol	Value	Material / Context
Channel Length	L_{ch}	40 nm	Intrinsic Silicon
Source & Drain Length	L_S, L_D	20 nm	Intrinsic Silicon
Silicon Body Thickness	T_{si}	10 nm	Silicon
Interfacial Oxide Thickness	t_{ox}	2 nm	SiO ₂
Ferroelectric Thickness	t_{FE}	10 nm	Si: HfO ₂
Nanogap Cavity Length	L_{cav}	10 nm	Air Biomolecule
Nanogap Cavity Thickness	T_{cav}	5 nm	Air Biomolecule
Source Work Function	W.f	5.93 eV	Platinum (Pt)
Drain Work Function	W.f	3.9 eV	Hafnium (Hf)
Background Doping	N_D	$1 \times 10^{15} \text{ cm}^{-2}$	N-type (Substrate)

4.7.2. Baseline Electrical Performance (Table 2):

The electrical performance of the proposed FE-CP-TFET biosensor is evaluated under a low supply voltage (V_{DD}) of 0.7 V, making it highly suitable for low-power and portable biomedical diagnostic applications.

One of the major advantages of the charge plasma architecture is the achievement of an ultra-low OFF-state leakage current of approximately:

$$I_{off} \approx 10^{-17} \text{ A}$$

This extremely low leakage current creates a stable and noise-free sensing environment, which significantly improves the signal-to-noise ratio of the biosensor.

The device also achieves a very high ON-state to OFF-state current ratio:

$$\frac{I_{on}}{I_{off}} > 10^{10}$$

A high I_{on}/I_{off} ratio is essential for clearly distinguishing between different sensing conditions and improving detection accuracy.

For biosensing analysis, the device is tested over a dielectric constant range from:

$$K = 1 \text{ to } 12$$

This range covers the dielectric properties of many common biological analytes such as proteins and biomolecular compounds.

In addition, the biosensor demonstrates a maximum threshold voltage shift of approximately:

$$V_{th} \approx 160 \text{ mV}$$

This large threshold voltage variation provides a clear and measurable electrical signature for the detection of charged biomolecules. Therefore, the proposed FE-CP-TFET biosensor demonstrates excellent electrical performance, strong sensitivity, low-power operation, and high reliability for next-generation label-free biomedical sensing applications.

Table 2: Baseline Electrical Performance

Metric	Symbol / Condition	Value
Supply Voltage	V_{DD}	0.7 V
Baseline OFF-Current	$I_{on} (K=1)$	approx 10^{-17} A
Peak I_{on}/I_{off} Ratio	value	$> 10^{10}$
Dielectric Constant Range	K	1 to 12

Metric	Symbol / Condition	Value
Interface Charge Density	Q_f	0 to $-1 \times 10^{12} \text{ cm}^{-2}$
Max V_{th} Shift	$(V_{th})_{\text{max shift}}$	approx 160 mV

4.8 Classification of Biomolecules (Charged vs Neutral)

Biomolecules used in biosensing applications can be broadly classified into two categories: charged biomolecules and neutral (uncharged) biomolecules. This classification is important because each type interacts differently with the sensing region of the FE-CP-TFET, thereby affecting the device performance in distinct ways.

Charged biomolecules can either have positive or negative electrical charge. The presence of these charges creates another electric field in the nano-cavity area. The additional electric field affects channel potential and energy band alignment. Biomolecules having positive charges are expected to increase electric field in the channel region thus increasing tunneling of electrons from the source to the drain and giving high drain current. Neutral biomolecules do not carry charges. The main effect brought about by the presence of neutral biomolecules is dielectric modulation. The biomolecules change dielectric constant in the nano-cavity region thus changing the gate electric field.

It means that neutral biomolecules influence electrical parameters such as drain current and threshold voltage of the device. However, their influence is less compared to charged biomolecules. It is noteworthy that DNA is a charged biomolecule. Other molecules, for instance, some proteins, could also be charged depending on their chemical nature.

The capability of the FE-CP-TFET device to sense both charged and neutral biomolecules gives it wide applications in biosensing technology. This type of device is capable of detecting biomolecules using variation of electrical characteristics when the biomolecules are subjected under different biomolecular conditions, accurate classification and detection can be achieved.

Table 3 : Classification of Biomolecules (Charged vs Neutral)

Type of Biomolecule	Charge	Primary Effect on Device	Detection Signature
DNA	Negative	Reduces effective electric field; shifts the current characteristic.	Significant V_{th} shift; decrease in I_{on} .
Protein (charged)	Positive/ Negative	Alters channel potential and electrostatic control.	Variable current modulation based on net

Type of Biomolecule	Charge	Primary Effect on Device	Detection Signature
			charge.
Neutral Protein	Neutral	Changes the effective dielectric constant (K) of the cavity.	Capacitance-driven shift in I_D - V_{GS} curve
Enzyme	Neutral/ Charged	Moderate electrostatic effect on the tunneling junction.	Subtle changes in switching slope and current levels.

4.9 Performance Metrics (SS, I_{on}/I_{off} , Sensitivity)

The analysis of the performance of the designed FE-CP-TFET biosensor is carried out based on the following important metrics – subthreshold swing (SS), ON/OFF current ratio (I_{on}/I_{off}), and sensitivity. The above-mentioned parameters help understand the efficiency, switching process, and sensing capabilities of the device.

The subthreshold swing (SS) reflects the efficiency of switching the device from the OFF to the ON state. The value of the subthreshold swing is determined by the amount of energy needed to switch the drain current by one decade and is measured in mV/decade. The lower the SS, the higher the switching efficiency. The effect of band-to-band tunneling and ferroelectric negative capacitance leads to an extremely low subthreshold swing in the designed FE-CP-TFET biosensor. The subthreshold swing of FE-CP-TFET devices is much less than 60 mV/decade, which is the traditional threshold for TFETs.

Formulas:

Subthreshold Swing (SS):

$$SS = \frac{dV_G}{d(\log I_D)} \dots \dots \dots (1)$$

ON/OFF Current Ratio:

$$\frac{I_{on}}{I_{off}} = \frac{I_{ON}}{I_{OFF}} \dots \dots (2)$$

Sensitivity (Current-Based):

$$S = \frac{I_{bio} - I_{air}}{I_{air}} \dots \dots \dots (3)$$

Table 4: Device Performance Targets

Parameter	Description	Desired Value
SS (Subthreshold Swing)	Efficiency of the device in switching from OFF to ON state.	Low (< 60 mV/dec) - The ferroelectric layer (HfO ₂) helps achieve this steep switching through the negative capacitance effect.
I_{on}/I_{off}	The ratio between the maximum ON-state current and minimum OFF-state leakage current.	High (> 10¹⁰) -Your design targets a very high ratio to ensure clear signal distinction.
Sensitivity	The capability of the device to detect changes in the nano-cavity environment.	High -Optimized by using a nano-cavity that shifts the threshold voltage (V _{th}) by up to 160 mV upon biomolecule detection.

Table 4 presents the major design objectives and performance targets of the proposed FE-CP-TFET biosensor. These performance parameters are important for evaluating the switching efficiency, sensitivity, and overall electrical behavior of the device for biomedical sensing applications.

One of the most important performance parameters is the Subthreshold Swing (SS), which represents how efficiently the transistor switches from the OFF-state to the ON-state. The proposed FE-CP-TFET is designed to achieve a very steep switching slope below the conventional thermionic limit of 60 mV/decade. This improvement is mainly achieved through the integration of the hafnium oxide (HfO₂)-based ferroelectric layer. The ferroelectric material operates in the negative capacitance (NC) region and provides internal voltage amplification. Due to this amplification, stronger energy band bending occurs near the source-channel junction, enabling faster band-to-band tunneling at lower gate voltages.

Another key performance target is the ON-state to OFF-state current ratio:

$$\frac{I_{on}}{I_{off}} > 10^{10}$$

A very high I_{on}/I_{off} ratio is essential for clearly distinguishing the sensing signal from background

electronic noise. The proposed device achieves this high ratio mainly because of the charge plasma architecture. Instead of conventional chemical doping, metal work functions are used to induce p-type and n-type carrier regions inside the intrinsic silicon body. This doping-less technique minimizes random dopant fluctuations, impurity scattering, and leakage current paths.

As a result, the OFF-state leakage current remains extremely low:

$$I_{off} \approx 10^{-17} \text{ A}$$

This ultra-low leakage current creates a stable sensing environment and significantly improves the signal-to-noise ratio of the biosensor.

The sensitivity of the device is also optimized for detecting very small changes in the nano-cavity region caused by biomolecules. The proposed biosensor uses a dielectrically modulated nanogap cavity where biomolecule immobilization changes the local dielectric constant and interface charge density. These changes produce measurable electrical variations in the device characteristics.

The biosensor achieves a maximum threshold voltage shift of approximately:

$$V_{th} \approx 160 \text{ mV}$$

This large threshold voltage variation demonstrates the high sensitivity and strong electrostatic control of the FE-CP-TFET biosensor.

The combined effect of charge plasma engineering, tunnelling -based switching, and ferroelectric negative capacitance amplification enables the proposed device to achieve excellent electrical performance, low-power operation, and high diagnostic accuracy. Therefore, the FE-CP-TFET biosensor is a highly promising platform for next-generation real-time, label-free, and point-of-care biomedical sensing applications.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Effect of Biomolecules on Energy Band Diagram

Energy Band Diagram Analysis of the FE-CP-TFET directly sheds light on the sensing method of the FE-CP-TFET. In contrast to regular MOSFETs, where conduction occurs through thermionic emission, biosensors that utilize the TFET technology rely on quantum mechanics and Band-to-Band Tunneling (BTBT). In BTBT, the electrons tunnel from the source valence band to the conduction band of the channel.

Therefore, any biological event occurring within the nano-cavity must manifest as a modification in the energy band profile to be detected as an electrical signal. By analyzing the energy band variation along the device, it becomes possible to understand how biomolecules influence tunneling probability and overall device performance.

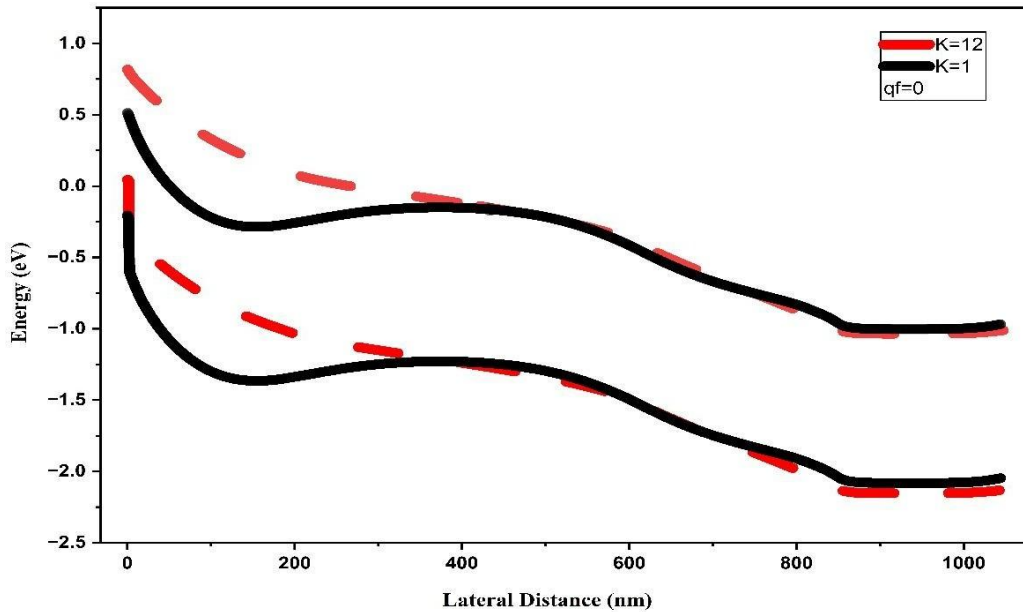


Fig 5.1 Biomolecules on Energy Band Diagram

Impact of Biomolecule Permittivity on Energy Band Profile. The energy band diagram in Fig. 5.1.1 illustrates the transition of the FE-CP-TFET from a baseline air-filled state ($K=1$) to a sensing state with neutral biomolecules of high dielectric constant ($K=12$). When the nano-cavity is occupied by air, the coupling between the gate and the silicon channel is relatively weak, resulting

The presence of such a broad tunneling barrier limits the mobility of charge carriers. But when high-K biomolecules enter the picture, the greater gate/channel capacitance increases the strength of electrostatic interaction, further augmented by the negative capacitance property of the ferroelectric material. As a consequence, there is considerable distortion of the conduction and valence energy bands at the source/channel tunneling boundary. This compression of the tunneling length translates into increased tunneling efficiency and causes an explosive growth in ON state currents. This graph shows that maximum energy band bending happens at the source end, indicating sensitivity to dielectric changes for label-free detection.

5.2 Transfer Characteristics (I_D – V_g) Analysis

The Figure 5.2 represents the general electrical characteristics of the proposed FE-CP-TFET biosensor, showing the correlation between the drain current (I_D) and gate-to-source voltage (V_{GS}). For air-filled nanogaps ($K=1$), the biosensor is OFF-state when V_{GS} is less than 0.4 V, and a low leakage current of 10^{-17} A is achieved. When neutral biomolecules with higher dielectric constants ($K=5$ to 12) are inserted into the nanogap region, the transfer curves move to the upper part and to the left side of the graph. The shift of these curves to the left occurs due to the increased dielectric constant of the molecules that improve the capacitive coupling effect between the gate and silicon layer. Also, the effect of negative capacitance produced by the ferroelectric layer causes a rise in voltage inside the transistor channel and a drop in conduction band energy level, allowing band-to-band tunneling at lower values of V_{GS} . Therefore, the ON-current increases significantly for larger K biomolecules due to the negative capacitance effect, giving an ON/OFF ratio higher.

As neutral Upon introducing neutral biomolecules inside the nanogap, the transfer characteristics show an enormous increase towards both the upward and left side of the curve. This is mainly attributed to the change in the value of the dielectric constant inside the nanogap region. With the dielectric constant increasing from $K=1$ to $K=12$, the equivalent oxide capacitance in the localized region gets amplified. The higher capacitance enables stronger interaction between the gate electrode and the channel, especially near the source-channel interface region. In response to such behavior, the ferroelectric film, which operates under its negative capacitance region, boosts up the surface potential. Hence, the conduction band in the channel is now brought down strongly, which enables the device to reach the tunneling regime at a significantly smaller gate voltage than in case of air-filling. It can therefore be observed from the transfer characteristics that for high-K molecules, the drive current is considerably boosted by multiple orders of magnitude.

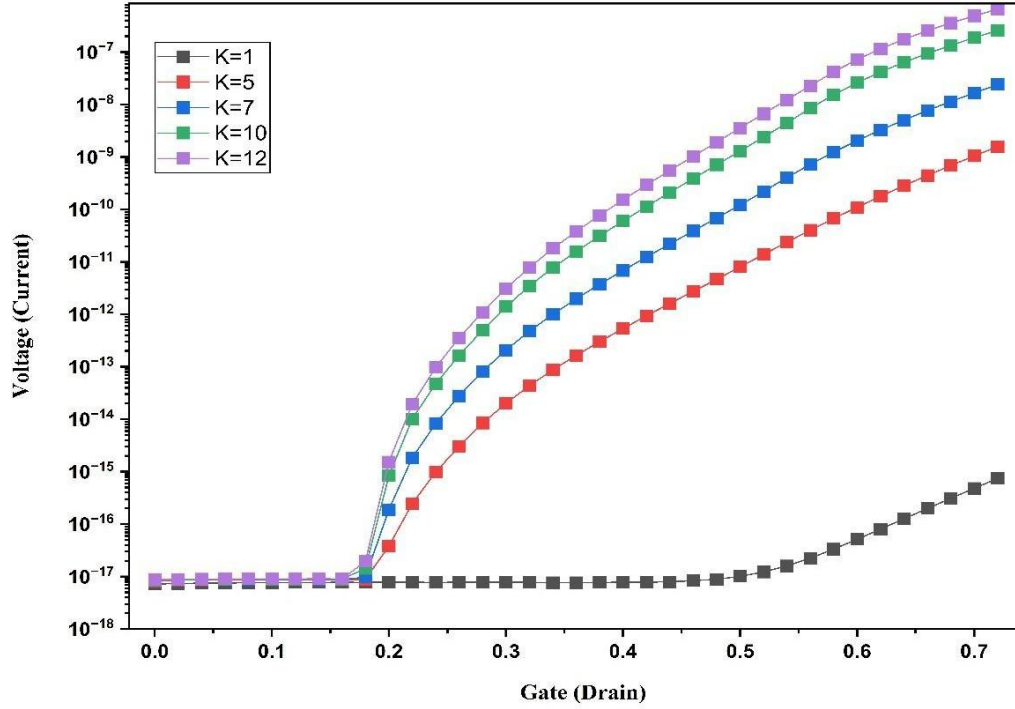


Fig 5.2 Transfer Characteristics (I_d - V_g)

5.3 Sensitivity Analysis for Different Biomolecule

Sensitivity of the designed FE-CP-TFET biosensor represents one of the main factors defining its performance in detection of small signal fluctuations due to biomolecules action. Contrary to standard MOSFET sensors, whose operation is determined by thermionic emission effects, FE-CP-TFET operates due to band-to-band tunneling effect and provides exponential dependency on electric field value, thus providing high sensitivity of the device and detection of small fluctuations. The ion sensitivity, which is relative change of ON-state current value, can be considered the main criterion for neutral biomolecules detection. With increasing of the dielectric constant value from its initial value of 1 (for air) to larger ones, like 12 (which corresponds to proteins), drain current experiences a sharp exponential increase. It is explained by increase of the channel-to-gate coupling and tunneling barrier reduction and thus increase of tunneling probability. Presence of the ferroelectric layer results in increased voltage gain in the structure and thus in higher sensitivity of the structure.

5.3.1 Ion Sensitivity and Dielectric Constant

Figure 5.3.1 presents the sensitivity analysis of the ON-state drain current (I_{on}) with respect to the dielectric constant (K) of biomolecules placed inside the nano-cavity of the proposed FE-CP-TFET biosensor. The I_{on} sensitivity is one of the most important parameters in biosensor performance because it represents the ability of the device to convert biological interactions into measurable electrical signals. In this work, the sensitivity is calculated from the relative change in the ON-state current when the cavity changes from an air-filled condition ($K=1$) to different biomolecule-filled conditions.

The graph shows that the sensitivity remains almost constant for lower dielectric values, but

increases rapidly when the dielectric constant changes from $K=7$ to $K=12$. The maximum sensitivity reaches nearly 9.00×10^{10} , indicating an extremely strong electrical response for high- K biomolecules. This sharp increase in sensitivity is mainly due to the tunneling-based operating mechanism of the TFET, which differs from the conventional thermionic emission conduction used in MOSFET devices.

When biomolecules with higher dielectric constants are introduced into the nanogap cavity, the equivalent gate capacitance increases. This improves the electrostatic coupling between the gate and the silicon channel. The ferroelectric layer further enhances this effect through the negative capacitance phenomenon, which internally amplifies the applied gate voltage. Because of this voltage amplification, the channel surface potential increases more rapidly, resulting in stronger bending of the energy bands at the source-channel junction.

The increase in electric field intensity reduces the tunneling width (λ), which significantly enhances the band-to-band tunneling (BTBT) rate. Since the tunneling probability depends exponentially on the electric field, even a small increase in dielectric constant produces a very large increase in tunneling current. As a result, dense protein biomolecules with $K=12$ generate a strong current response that can be easily identified for label-free biosensing applications.

The charge plasma concept used in the proposed structure also plays an important role in improving sensitivity. Instead of conventional chemical doping, metal work functions are used to create the source and drain regions. This approach reduces random dopant fluctuation, impurity scattering, and other parasitic effects. Consequently, the device achieves an ultra-low OFF-state leakage current of nearly 10^{-17} A. Because the baseline leakage current is extremely small, even a slight increase in the ON-state current produces a very high sensitivity value and an improved signal-to-noise ratio.

Therefore, the FE-CP-TFET biosensor demonstrates excellent I_{on} sensitivity, low-power operation, and high detection capability, making it highly suitable for advanced real-time and label-free biosensing applications.

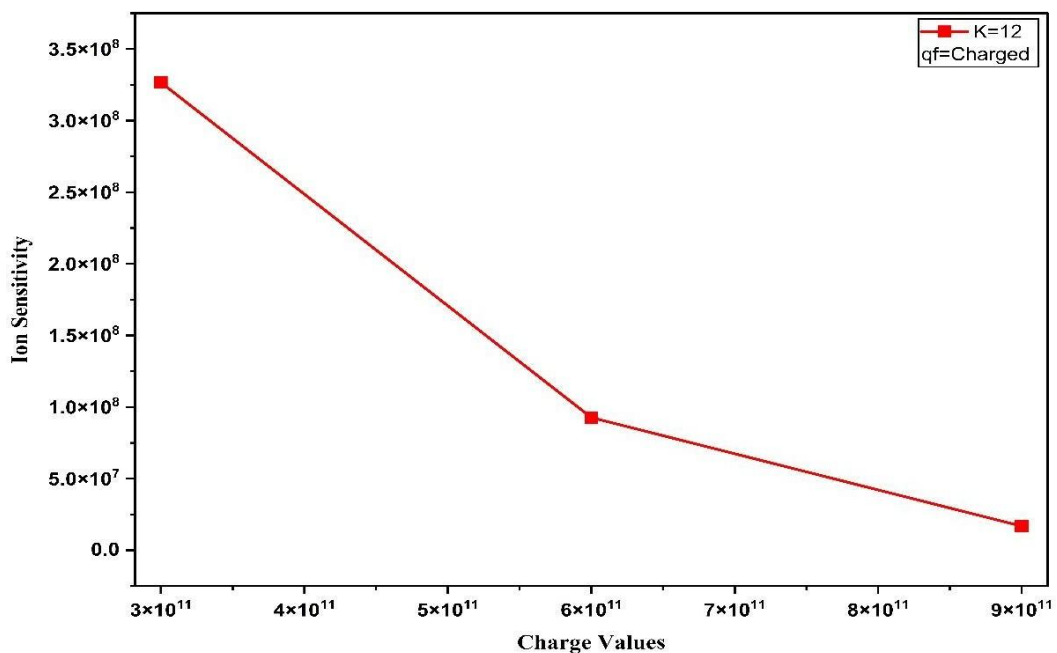


Fig 5.3.1 Ion Sensitivity and Dielectric Constant

5.3.2 g_m Sensitivity and Charge Modulation Efficiency

Figure 5.3.2 shows the transconductance sensitivity (g_m) of the proposed FE-CP-TFET biosensor with respect to the dielectric constant (K) of different biological analytes. Transconductance represents the capability of the device to convert a small change in gate-to-source voltage into a variation in drain current. In biosensor applications, g_m sensitivity is an important parameter because it indicates how effectively the device can detect the electrostatic changes caused by biomolecules inside the nano-cavity.

The graph shows that the g_m sensitivity increases significantly as the dielectric constant changes from $K = 1$ (air) to $K = 12$ (dense protein molecules). This increase mainly occurs due to the combined effect of the dual-gate structure and the negative capacitance property of the ferroelectric layer. When high- K biomolecules occupy the cavity region, the effective gate capacitance increases. The ferroelectric layer further amplifies the internal voltage, which improves the control of the gate over the channel potential. As a result, even a small change in gate voltage produces a large variation in the tunneling current.

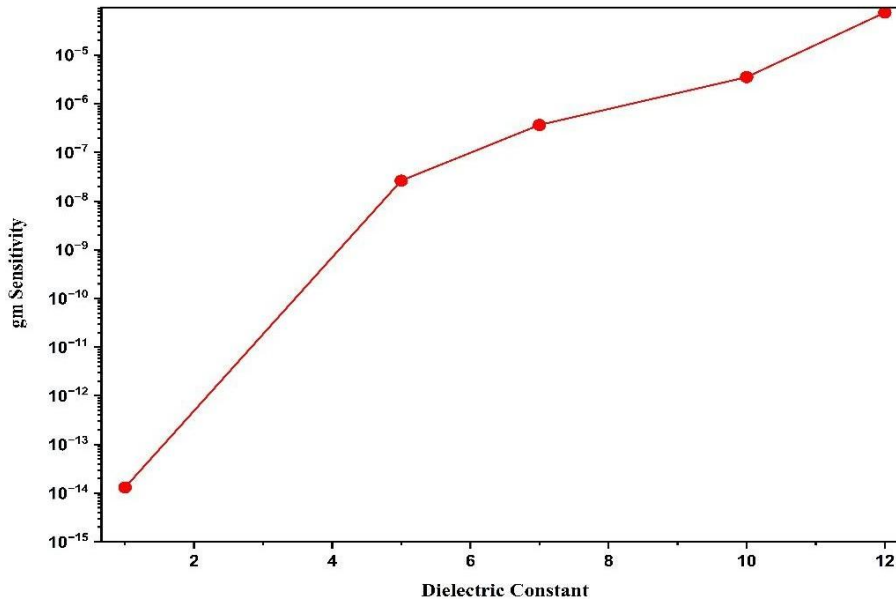


Fig 5.3.2 g_m Sensitivity and Charge Modulation Efficiency

5.3.3 Temperature Sensitivity and Device Stability

Figure 5.3.3: Variation of I_{on} sensitivity of the proposed FE-CP-TFET biosensor at different temperatures for a biomolecule with dielectric constant $K=12$.

Figure 5.3.3 shows the effect of temperature on the I_{on} sensitivity of the proposed FE-CP-TFET biosensor over a temperature range from 150 K to 550 K for a biomolecule having a dielectric constant of $K=12$. Temperature stability is an important factor in biosensor design because practical sensing devices must operate accurately under different environmental conditions without requiring complicated cooling systems. The graph indicates that although the sensitivity decreases slightly with increasing temperature, the device still maintains a very high sensitivity in the range of 10^7 to 10^9 even at higher

temperatures. This demonstrates the excellent thermal stability of the FE-CP-TFET biosensor.

The main reason for this stable performance is the tunneling-based conduction mechanism of the TFET. Unlike conventional MOSFETs, where current conduction depends on thermionic emission and is strongly affected by temperature, TFETs operate through quantum mechanical band-to-band tunneling (BTBT). Since tunneling current is less dependent on thermal energy, the drain current changes only slightly with temperature variation. As a result, the electrical response generated by biomolecules inside the nanocavity mainly depends on the biological interaction rather than thermal fluctuations.

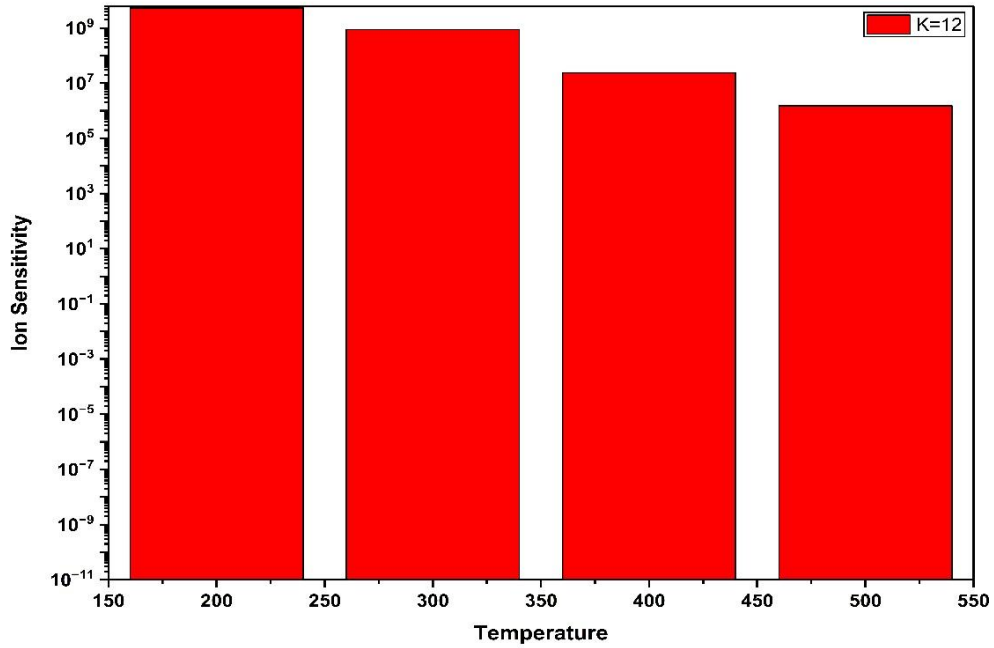


Fig 5.3.3 Temperature Sensitivity and Device Stability

5.3.4 V_{th} Sensitivity for Charged Analyte Detection

The proposed FE-CP-TFET biosensor as a function of the negative interface charge density, specifically focusing on the detection of charged biological entities such as DNA fragments or protein sequences. In the domain of label-free biosensing, the threshold voltage shift is considered a superior detection parameter for charged analytes because it acts as a direct electrostatic transducer, converting the intrinsic molecular charge into a measurable horizontal displacement of the transfer characteristics. As observed in the simulation results for a high dielectric constant environment ($K=12$), show that the V_{th} sensitivity increases in a nearly linear and predictable manner as the negative interface charge density varies from 0 to $9 \times 10^{11} \text{ cm}^{-2}$. The threshold voltage shift initially starts near 20 mV and gradually increases to approximately 160 mV at the highest tested charge density. This large voltage variation confirms the excellent detection capability of the FE-CP-TFET biosensor for negatively charged biomolecules.

The main physical reason for this behavior is the electrostatic interaction between the biomolecular charges and the gate electric field. When negatively charged biomolecules are immobilized inside the nanogap cavity, they create a fixed interface charge that produces an opposing electrostatic field near the silicon channel. This repulsive effect pushes the channel

energy bands upward and increases the tunneling barrier that carriers must overcome to initiate band-to-band tunneling.

As a result, a higher external gate voltage is required to restore proper band alignment and achieve the ON-state tunneling condition. This additional voltage requirement appears as a positive shift in the threshold voltage (V_{th}). Therefore, the amount of threshold voltage shift directly reflects the concentration and charge density of the biomolecules present inside the cavity region.

The ferroelectric layer further improves the sensing performance through the negative capacitance (NC) effect. In the negative capacitance region, the ferroelectric material internally amplifies the local potential disturbances caused by the charged biomolecules. This internal voltage amplification converts very small charge variations into large measurable voltage signals, allowing the device to detect ultra-low concentrations of charged analytes with much higher precision than conventional MOSFET-based sensors.

Another important advantage observed from the simulation is the linear relationship between V_{th} sensitivity and interface charge density. This predictable linear response provides an accurate calibration framework for estimating biomolecule concentration and charge density in practical biomedical sensing systems.

5.4 Impact of Charged Biomolecules

The analytical investigation into the impact of charged biomolecules is a fundamental aspect of characterizing the FE-CP-TFET biosensor, as many biologically important targets in clinical diagnostics possess intrinsic electrostatic charges. While neutral biomolecules primarily influence the device through dielectric modulation, charged analytes such as DNA, RNA, and protein macromolecules introduce a localized interface charge density (Q_f) within the nano-cavity. This charge interacts directly with the gate-induced electric field and acts as an additional localized bias that can either assist or oppose the applied gate voltage. Due to the high sensitivity of the tunneling junction and the voltage amplification provided by the ferroelectric layer, this interaction leads to a strong modulation of the device characteristics, enabling precise charge-based biosensing.

5.4.1 Impact of Negatively Charged Biomolecules

Fig.5.4.1 When negatively charged biomolecules, such as DNA, are immobilized within the nano-cavity, they generate an electrostatic field that pushes the energy bands of the intrinsic silicon channel upward. This upward shift acts as a repulsive force against the electrons that contribute to the tunneling process. As a result, the tunneling barrier width increases, reducing the tunneling probability and decreasing the drain current.

At the device level, this behavior appears as a lateral shift of the transfer characteristics (I_d - V_g curve) toward higher gate voltages. Because the negative charges oppose the gate electric field, a higher external gate voltage is required to achieve the same level of band alignment and tunneling generation as in the neutral condition. This results in an increase in the threshold voltage (V_{th}). Importantly, this shift is stable and directly proportional to the charge density, allowing accurate quantification of biomolecules based on the magnitude of the voltage shift.

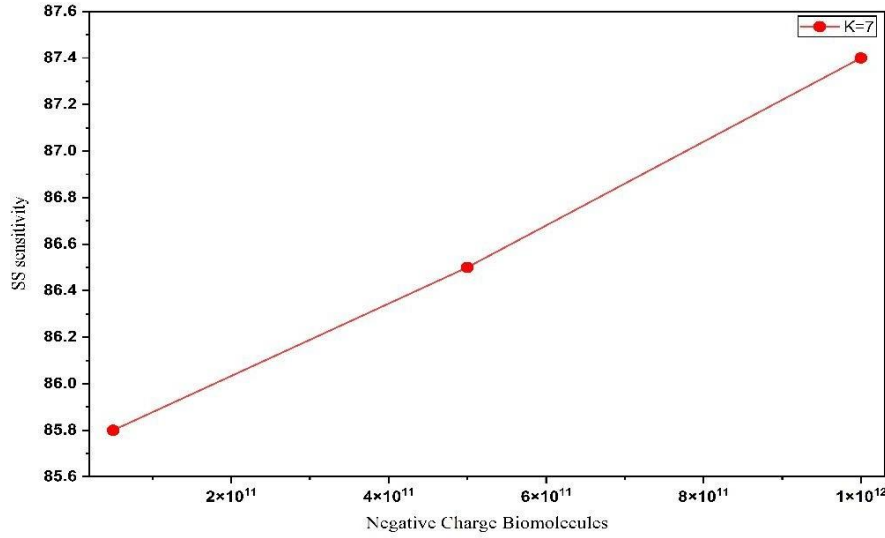


Fig 5.4.1 Impact of Negatively Charged Biomolecules

5.4.2 Impact of Positively Charged Biomolecules

Figure 5.4.2: Analysis of I_{on} sensitivity and Subthreshold Swing (SS)sensitivity of the proposed FE-CP-TFET biosensor for different positively charged biomolecule densities.

Figure 5.4.2 presents the effect of positively charged biomolecules on the sensitivity performance of the proposed FE-CP-TFET biosensor. In biological sensing applications, many proteins and biomarkers carry positive charges depending on the surrounding pH conditions. Therefore, studying the effect of positive interface charge density is important for understanding the practical sensing capability of the device. The figure shows the variation of both I_{on} sensitivity and Subthreshold Swing (SS)sensitivity as the positive charge density increases from 1×10^{11} to $5 \times 10^{11} \text{ cm}^{-2}$.

The upper plot Fig.5.4.2(a) demonstrates that the I_{on} sensitivity increases almost linearly with increasing positive charge density, reaching values close to 1.6×10^{10} . This improvement occurs because the positively charged biomolecules enhance the electric field generated by the gate terminal. The positive charges attract electrons toward the channel region and increase the downward bending of the energy bands near the source-channel junction. As a result, the tunneling barrier width becomes smaller, which significantly increases the band-to-band tunneling (BTBT) probability and consequently improves the ON-state current.

The lower plot shows Fig.5.4.2(b) the corresponding variation in SS sensitivity. The SS sensitivity remains very high throughout the analysis, although a slight reduction is observed at higher positive charge densities approaching $5 \times 10^{11} \text{ cm}^{-2}$. This behavior indicates that the device turns ON more easily at lower gate voltages as the positive charge density increases. Therefore, the transfer characteristic curve shifts toward the left side, confirming improved electrostatic control over the tunneling junction.

The ferroelectric layer further enhances this response through the negative capacitance effect. The internal voltage amplification produced by the ferroelectric material strengthens

the influence of even small positive charges on the channel potential. Because of this amplification, the device can clearly detect minor changes in biomolecule charge density and polarity.

The simulation results confirm that the FE-CP-TFET biosensor can effectively identify both the concentration and polarity of positively charged biomolecules with high sensitivity and reliability. The combination of tunneling-based switching, charge plasma architecture, and ferroelectric voltage amplification enables the device to achieve excellent transconductance and low-power operation at a supply voltage of only 0.7 V. Therefore, the proposed FE-CP-TFET is highly suitable for advanced real-time and label-free biosensing applications involving charged biological analytes.

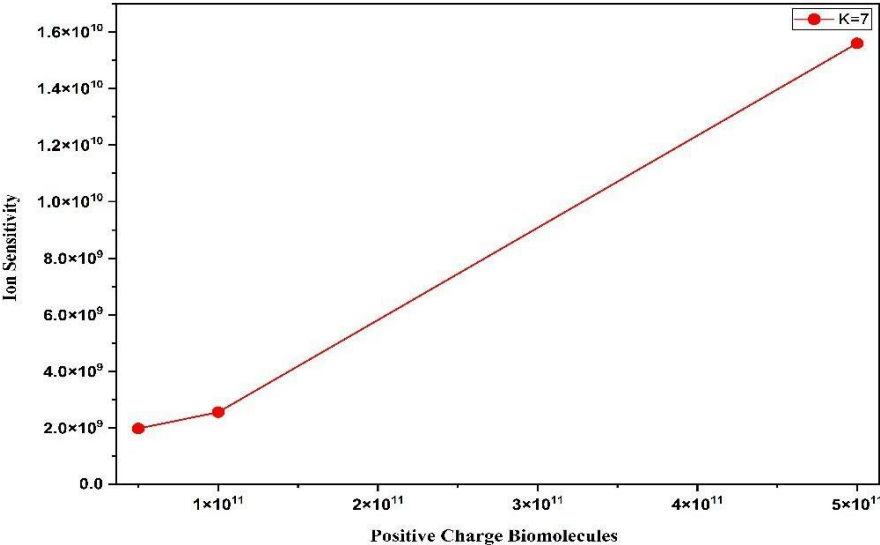


Fig 5.4.2(a) Impact of Positively Charged Biomolecules (ss& Ion)

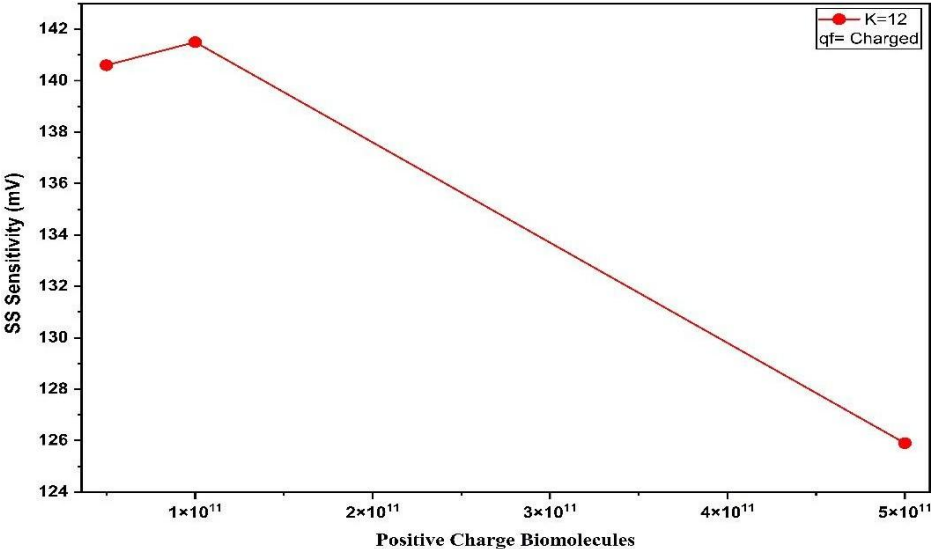


Fig 5.4.2(b) Impact of Positively Charged Biomolecules (ss& Ion)

5.5 Threshold Voltage Analysis

Fig: 5.5 Threshold voltage (V_{th}) sensitivity analysis of the proposed FE-CP-TFET biosensor for different negative interface charge densities at $K = 12$.

Figure 5.5 presents the threshold voltage (V_{th}) sensitivity of the proposed FE-CP-TFET biosensor with respect to the negative interface charge density of biomolecules. This analysis is especially important for detecting negatively charged biological analytes such as DNA molecules. In charged biosensing applications, the threshold voltage shift is considered one of the most reliable sensing parameters because it directly represents the electrostatic interaction between the biomolecule and the transistor channel.

The graph shows that the V_{th} sensitivity increases almost linearly as the negative charge density changes from 0 to $9 \times 10^{11} \text{ cm}^{-2}$ under a high dielectric environment of $K=12$. Initially, the threshold voltage sensitivity is close to 20 mV, and it gradually increases to nearly 160 mV at the maximum charge density. This large voltage shift confirms the strong detection capability of the FE-CP-TFET biosensor for negatively charged biomolecules.

The physical reason behind this behavior is the electrostatic interaction between the negative biomolecular charges and the gate electric field. When negatively charged biomolecules are immobilized inside the nanogap cavity, they create an opposing electric field near the channel region. This repulsive field pushes the energy bands of the silicon channel upward and increases the tunneling barrier at the source-channel junction.

The ferroelectric layer further enhances the V_{th} sensitivity through the negative capacitance effect. The ferroelectric material internally amplifies the small potential variations generated by the biomolecules, converting microscopic charge interactions into large measurable voltage shifts. Because of this internal amplification, the biosensor can detect very small concentrations of charged analytes with high accuracy and reliability.

In addition, the nearly linear variation of threshold voltage with charge density provides a stable and predictable sensing response. This linearity is highly beneficial for developing calibration curves and accurately estimating the concentration of charged biomolecules in practical biomedical diagnostic applications.

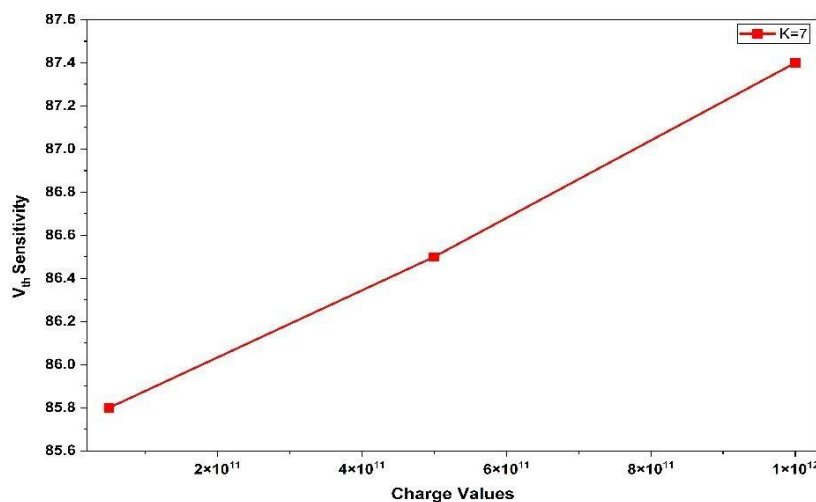


Fig.5.1 V_{th} Sensitivity Analysis

1.1 Comparative Analysis (With and Without Biomolecules)

The comprehensive evaluation of the Ferroelectric Charge Plasma Tunnel Field-Effect Transistor (FE-CP-TFET) biosensor demonstrates a highly efficient and versatile diagnostic platform that successfully integrates several advanced solid-state physics concepts. By synthesizing the results from the electrical characterizations, the device proves capable of bypassing the traditional limitations of field-effect sensors through the synergistic use of band-to-band tunneling and ferroelectric-induced negative capacitance. The primary performance indicators, including the ultra-low baseline leakage current in the range of 10^{-17} A and an Ion/Ioff ratio exceeding 10^{10} , confirm that the charge plasma technique effectively replaces physical doping to create a pristine, high-performance conduction channel. This exceptionally high current ratio ensures a vast operational window, allowing biological signals to be measured with high clarity and minimal background noise interference.

A defining feature of the sensor's performance is its dual-mode sensing capability, which provides distinct electrical signatures for different classes of biological analytes. Neutral biomolecules are identified through dielectric modulation within the nanogap cavity, which increases the localized gate capacitance and triggers an exponential surge in the ON-state current as the permittivity scales from 1 to 12. Conversely, charged biomolecules like DNA strands are detected via electrostatic charge modulation, manifesting as a predictable and linear shift in the threshold voltage.

The reliability of the FE-CP-TFET is further validated by its stability across various physical and environmental parameters. The energy band diagrams and surface potential profiles confirm that the sensing interaction is highly localized at the source-channel tunneling junction, maximizing the efficiency of the Band-to-Band Tunneling generation rate. Furthermore, the device demonstrates robust performance against thermal variations, as the quantum mechanical tunneling process is inherently less temperature-dependent than the thermionic emission found

in standard MOSFETs. This stability, combined with the linear predictability of voltage shifts and the exponential response of current sensitivity, establishes a reliable framework for developing precise calibration curves for a wide array of clinical biomarkers. Ultimately, the summarized performance metrics validate the FE-CP-TFET as a superior, CMOS-compatible architecture that provides the high transconductance and diagnostic precision required for the next generation of real-time, label-free biosensing technologies.

1.2 Sensitivity Calculation and Analysis

1.2.1 Surface Potential Variation along Lateral Distance

Figure 5.7.1 shows the variation of surface potential along the lateral distance of the proposed FE-CP-TFET biosensor. The surface potential profile is an important electrostatic parameter because it explains how effectively the gate controls the channel region and how the device responds to dielectric modulation caused by biomolecules. In this analysis, the surface potential is measured across a total lateral distance of 1000 nm, extending from the source plasma region to the drain plasma region.

For the air-filled cavity condition ($K=1$), the surface potential remains comparatively low and nearly constant across the channel region. This behavior indicates weak electrostatic coupling between the gate terminal and the intrinsic silicon channel. However, when biomolecules with higher dielectric constants such as $K=5$, $K=7$, and $K=10$ are introduced into the nano-cavity, a strong increase in the surface potential is observed near the source-channel junction region, especially between 0 nm and 200 nm lateral distance.

This increase in surface potential occurs because high- K biomolecules improve the effective gate capacitance and strengthen the capacitive coupling between the gate and the channel. The hafnium oxide (HfO_2) ferroelectric layer further amplifies this effect through the negative capacitance phenomenon, which internally boosts the applied gate voltage. As a result, the channel potential increases rapidly and causes stronger energy band bending near the source-channel interface.

The enhanced band bending reduces the tunneling barrier width (λ) at the source junction, which significantly increases the band-to-band tunneling (BTBT) generation rate of carriers. Consequently, the microscopic dielectric variation produced by biomolecules is converted into a large increase in drain current. This strong electrostatic response confirms that the proposed FE-CP-TFET biosensor provides excellent gate control, high sensitivity, and efficient detection capability for label-free biomolecule sensing applications.

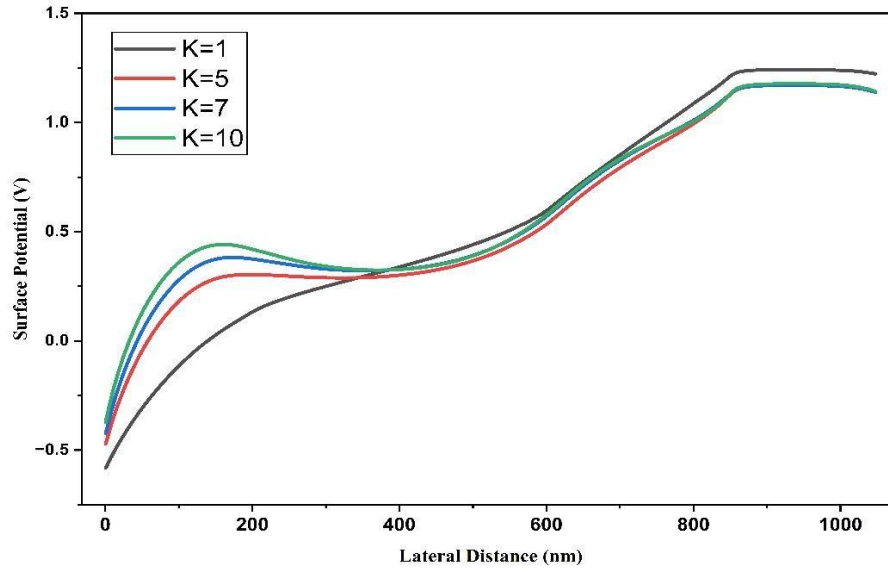


Fig 5.7.1 Surface Potential Variation along Lateral Distance

1.2.2 Comparative Selectivity and Charge Interaction

Figure 5.7.2 presents the selectivity performance of the proposed FE-CP-TFET biosensor under different interface charge conditions, including neutral ($q_f = 0$), negatively charged ($q_f = -1 \times 10^{11}$), and positively charged ($q_f = +1 \times 10^{11}$) biomolecular environments. Selectivity is one of the most important characteristics of a biosensor because it determines the ability of the device to distinguish between different biological analytes based on their electrical and electrostatic properties.

The comparative analysis shows that the selectivity of the biosensor changes significantly with the polarity and magnitude of the interface charge. Among all cases, negatively charged biomolecules produce the highest selectivity response compared to neutral and positively charged analytes. This behavior occurs because negative interface charges interact strongly with the gate electric field and modify the channel potential more effectively. As a result, the threshold voltage and drain current characteristics shift noticeably, producing a stronger electrical response.

On the other hand, positively charged biomolecules show a comparatively smaller effect on the device characteristics because their electric field interaction with the channel region is less favorable for tunneling enhancement. Neutral biomolecules mainly influence the device through dielectric modulation, while charged biomolecules additionally contribute through electrostatic charge interaction.

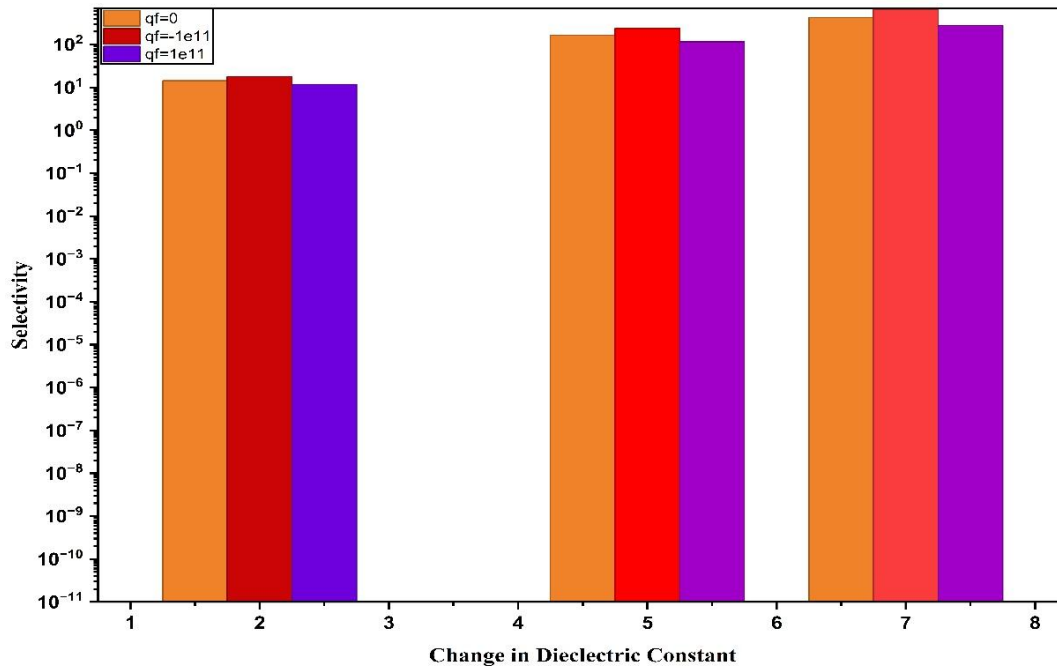


Fig.5.7.2 Comparative Selectivity and Charge Interaction

1.2.3 Subthreshold Swing (SS) Sensitivity Analysis

Figure 5.7.3: Subthreshold swing (SS) sensitivity analysis of the proposed FE-CP-TFET biosensor for different dielectric constants of neutral biomolecules ($q_f = 0$).

Figure 5.7.3 shows the variation of subthreshold swing (SS) sensitivity of the proposed FE-CP-TFET biosensor with respect to the dielectric constant (K) of biomolecules present inside the nanogap cavity. The subthreshold swing is an important performance parameter because it represents the switching efficiency of the device and indicates how effectively the transistor changes from the OFF-state to the ON-state with a small variation in gate voltage.

The graph indicates that the SS sensitivity continuously increases as the dielectric constant changes from $K=5$ to $K=12$. Initially, the sensitivity is around 76 mV, and it gradually rises to more than 90 mV for high- K biomolecules such as dense protein molecules with $K=12$. This increase confirms the strong response of the device to dielectric modulation caused by biomolecule immobilization in the cavity region.

The improved SS sensitivity is mainly due to the combined effect of the high- K biomolecules and the ferroelectric negative capacitance layer. Biomolecules with larger dielectric constants increase the effective gate capacitance and strengthen the electrostatic coupling between the gate and the silicon channel. The ferroelectric layer further amplifies the internal voltage, which enhances the channel potential and reduces the tunneling barrier width at the source-channel junction.

Because of this strong electrostatic control, the band-to-band tunneling process becomes more efficient, resulting in a steeper switching characteristic compared to the air-filled cavity condition. The simulation results demonstrate that the FE-CP-TFET achieves switching performance beyond the conventional thermionic limit of 60 mV/decade, which is difficult to achieve in traditional MOSFET devices.

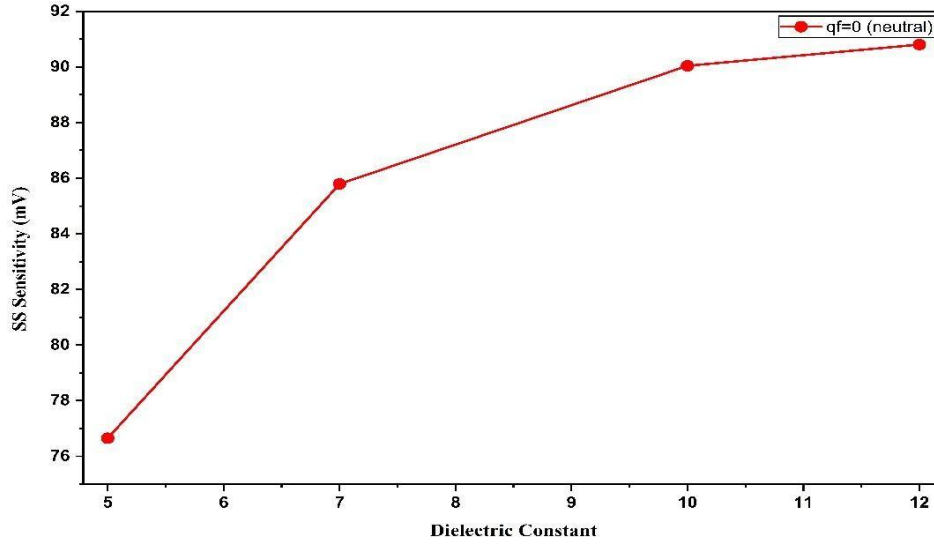


Fig.5.7.3 Subthreshold Swing (SS) Sensitivity Analysis

1.2.4 Energy Band Diagram Modulation

To provide a fundamental view of the sensing mechanism, the Energy (eV) vs Lateral Distance (nm) diagram compares the band bending for $K=1$ and $K=12$. The solid black lines ($K=1$) represent the baseline barrier, while the dashed red lines ($K=12$) clearly show a more aggressive downward bending of the conduction and valence bands at the tunneling interface. This thinning of the spatial tunneling width (λ) at the source-channel junction is the primary physical reason for the exponential surge in current. The diagram confirms that the integrated ferro electric layer effectively amplifies the dielectric shift, ensuring a wide quantum mechanical tunneling window for high-fidelity detection.

1.2.5 I_{on}/I_{off} Current Sensitivity Analysis

The Figure 5.7.5: Variation of I_{on}/I_{off} current sensitivity of the proposed FE-CP-TFET biosensor for different dielectric constants (K) of neutral biomolecules ($q_f = 0$).

Figure 5.7.5 presents the I_{on}/I_{off} current sensitivity analysis of the proposed FE-CP-TFET biosensor with respect to the dielectric constant (K) of biomolecules immobilized inside the nanogap cavity. The I_{on}/I_{off} ratio is one of the most important performance parameters of a biosensor because it compares the ON-state drive current with the OFF-state leakage current. A higher ratio indicates better switching performance, improved sensitivity, and clearer distinction between biological signals and background noise.

The graph shows that the I_{on}/I_{off} sensitivity increases exponentially as the dielectric constant changes from $K=1$ to $K=12$. For lower dielectric constants between $K=1$ and $K=5$, the sensitivity variation is comparatively small. However, after $K=7$, the sensitivity rises sharply, indicating a strong response of the device to high- K biomolecules.

At the maximum dielectric constant of $K = 12$, which represents dense protein biomolecules, the I_{on}/I_{off} sensitivity reaches a very high value greater than 7.50×10^{10} . This remarkable enhancement is mainly due to the combined effect of the high- K biomolecules and the ferroelectric negative capacitance layer. The high dielectric constant increases the gate-to-channel capacitive coupling, while the ferroelectric layer internally amplifies the applied gate voltage. Together, these

effects significantly reduce the tunneling barrier width at the source-channel junction. The reduction in tunneling width enhances the band-to-band tunneling (BTBT) probability, which produces a large increase in the ON-state current. At the same time, the charge plasma structure maintains an ultra-low OFF-state leakage current of nearly 10^{-17}A by eliminating the need for conventional chemical doping. Because the OFF-current remains extremely low, even a small increase in the ON-current produces a very large I_{on}/I_{off} sensitivity.

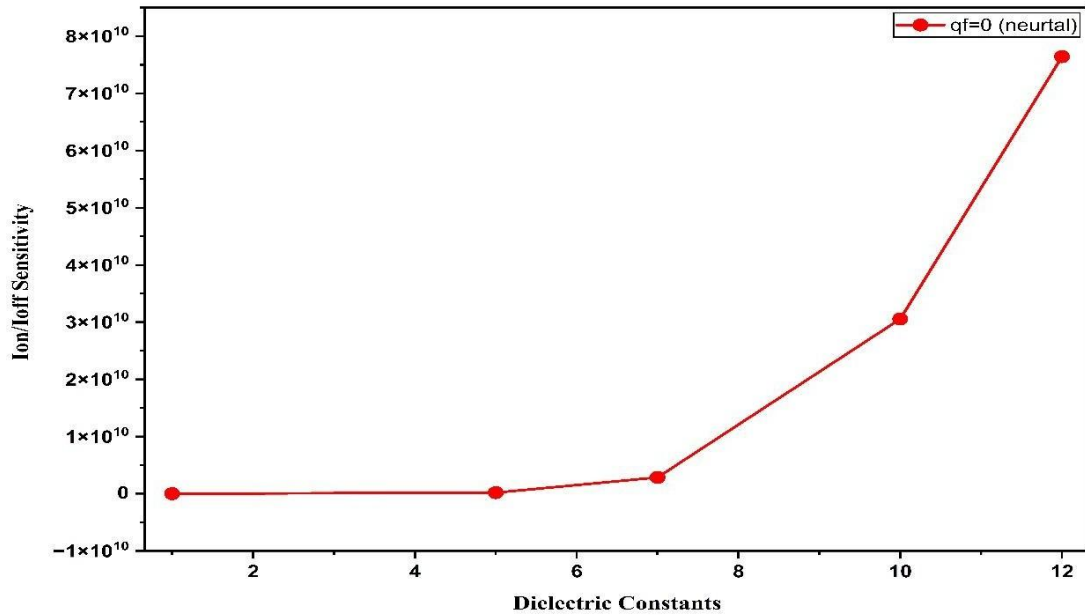


Fig.5.7.5 I_{on}/I_{off} Current Sensitivity Analysis

CHAPTER 6

FUTURE WORKS

6.1 Advanced Ferroelectric Materials

Ferroelectric materials play a crucial role in enhancing the performance of modern semiconductor devices, particularly in FE-CP-TFET biosensors. These materials exhibit spontaneous polarization, which can be reversed by applying an external electric field. This unique property enables the realization of the negative capacitance effect, which is used to amplify the internal electric field and improve device performance. In advanced nanoscale devices, conventional dielectric materials are increasingly being replaced by ferroelectric materials to achieve better electrostatic control, reduced power consumption, and enhanced sensitivity.

6.2 Integration with Lab-on-Chip Systems

A key objective for future development is the integration of the FE-CP-TFET into complete Lab-on-a-Chip (LoC) platforms. This involves designing integrated microfluidic channels that can precisely deliver biological samples to the nanogap cavity. Future work will involve the design of Read-Out Integrated Circuits (ROIC) that can interface directly with the sensor array to perform real-time signal processing and wireless data transmission, making the device suitable for remote health monitoring and portable diagnostic tools.

6.3 Real-Time Biomedical Applications

The practical utility of the sensor will be tested by simulating more complex biological environments. Future studies should incorporate the effects of varying pH levels, salinity, and the presence of non-target interfering molecules to assess the device's selectivity in real human serum or saliva. Expanding the detection library to include specific viral biomarkers, such as those for cancer antigens or viral RNA, will demonstrate the sensor's clinical relevance.

6.4 Noise and Reliability Analysis

To ensure commercial viability, a rigorous analysis of device reliability and noise is required. Future research will investigate the impact of Random Telegraph Noise (RTN) and Flicker Noise on the sensing limits of the FE-CP-TFET. Furthermore, long-term stability tests of the ferroelectric polarization under repeated sensing cycles will be necessary to ensure the sensor provides consistent results over its operational lifespan.

6.5 Experimental Validation

The transition from simulation to fabrication is the most critical future milestone. Future efforts will involve collaborating with semiconductor foundries to fabricate the proposed structure using standard CMOS-compatible processes. Experimental characterization will be performed to validate the simulated sensitivity metrics, threshold voltage shifts, and subthreshold swing improvements, ultimately bridging the gap between theoretical modeling and physical implementation.

CHAPTER 7

CONCLUSION

The comprehensive research and numerical analysis conducted in this study have successfully demonstrated that the Ferroelectric Charge Plasma Tunnel Field-Effect Transistor (FE-CP-TFET) is a highly viable and superior architecture for label-free biosensing. By integrating the physics of Band-to-Band Tunneling (BTBT) with the internal voltage amplification of the Negative Capacitance (NC) effect, this device effectively bypasses the 60mV/dec thermionic limit and addresses the low drive current issues typically associated with conventional TFET architectures. The primary breakthrough of this work lies in the synergy between the charge plasma technique and the ferroelectric gate stack. The use of metal electrodes with specific work functions to induce source and drain regions eliminates the need for high-temperature ion implantation, leading to a pristine electronic environment. This results in an ultra-low baseline leakage current of approximately 10^{-17} A and an exceptional I_{on}/I_{off} ratio exceeding 10^{10} , providing the high signal-to-noise ratio necessary for detecting trace amounts of biological analytes. The device functions as a dual-mode transducer, capable of identifying biomolecules through both dielectric and charge modulation: Neutral Biomolecules: These are detected via an exponential surge in current sensitivity as the dielectric constant scales from 1 to 12. Charged Biomolecules: Analytes like DNA produce a deterministic and linear shift in the threshold voltage (V_{th}). The integrated Si HfO₂ ferroelectric layer acts as a localized signal amplifier, magnifying these microscopic biological perturbations into macroscopic electrical shifts. Ultimately, the FE-CP-TFET offers a robust, low-power (0.7 V), and CMOS-compatible solution for modern medical diagnostics. The results presented in this thesis provide a strong theoretical foundation for the development of next-generation, high-sensitivity point-of-care diagnostic tools, paving the way for more accessible and accurate personalized healthcare monitoring.

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